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RESEARCH ARTICLE

A Hybrid LSTM-GRU Model for Stock Price Prediction

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ABSTRACT The dynamic variation of the stock market plays a crucial role in assessing a country's economic power and development. Modeling the chaotic fluctuations in stock prices aids investors and traders in uncertain situations by evaluating market trends for investment decisions. Previous methods utilized Recurrent Neural Network (RNN) models for prediction. However, none of the previous works attempted to leverage the full range of technical analysis features and use Gated Recurrent Unit (GRU) and Long Short-Term Memory (LSTM) modules to capture short and long dependencies. Initially, we use machine learning to extract appropriate features for making predictions. Subsequently, the impact of each feature is adjusted proportionally to the weight matrix extracted from the neural network. We combined LSTM and GRU to capture long and short information horizons in stock sequences to mitigate limitations of prior methods, such as the difficulty in hyperparameter tuning. By combining the capabilities of these two networks along together, we present a model for predicting market fluctuations. Our proposed model was evaluated in a specified test market, achieving a 3% improvement in Mean Squared Error (MSE) over the previous best approaches.

INDEX TERMS Stock prediction, RNN, LSTM, GRU, hybrid approach.

I. INTRODUCTION

The stock market is one of the most important sources of revenue for companies. The stock market allows businesses to trade publicly or expand their company by selling their stock in the public market. History has indicated that the price of stocks and other assets is an essential part of the dynamics of economic activity and can affect social mood indicators or be affected by them. The stock market is often the primary indicator of a country's economic power and development. An increase in stock prices is associated with increased investment in the business and vice versa. Stock prices also influence household wealth and consumption; therefore, predicting changes in the stock market's future is important for companies and investors in the financial market. In recent years, the use of social media has expanded significantly, and the influence of users on the

stock market has increased. Moreover, market sentiment and investor behavior are cardinal for detecting market trends and stock prices in these systems. Sentiment analysis can help gain valuable insights into the overall mood of the market by identifying sentiment in news, reports, or social media [1]. In recent years, machine learning (ML) has been the main solution to solve complex problems among different domains, and researchers have striven to solve different problems using ML. Accurate stock market prediction is essential for global finance. It empowers investors to make informed decisions, minimize risk, and maximize returns while assisting policymakers in understanding economic trends and stabilizing the economy. Given its influence on global trade, investment, and economic policy, stock market prediction plays a vital role in the interconnected global financial system. This study focuses on improving stock price prediction within the Tehran Stock Exchange (TSE), a specific market with unique characteristics that is less extensively studied in the international literature compared

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to major global financial centers. Our core contribution is the proposal and empirical evaluation of a hybrid LSTM-GRU deep learning architecture combined with a comprehensive set of 42 technical indicators. This specific architectural combination leverages the respective strengths of GRU and LSTM networks to capture both short-term fluctuations and long-term dependencies in sequential financial data, while the technical features provide a rich representation of market dynamics. Therefore, predicting stock market trends using social media analysis can be practical. Researchers increasingly use learning models due to the availability of big data. The ability of deep learning models to automatically select features has dramatically increased the adoption of deep learning approaches over the last decade. Valuation and technical analysis are used to improve prediction. Characteristics of a stock index, such as opening price, closing price, minimum price, and maximum price, are insufficient for accurate predictions. Many researchers have proposed numerous technical approaches to generate feature spaces to address this. Many of these techniques use opening and closing prices to build new indicators. In the work by Sivadasan et al., the effect of the indicators on different architectures of deep learning for time series prediction has been studied [2]. This study aims to use technical indicators to create a rich feature space for the Tehran stock market prediction. The feature selection for stock prediction has not been established in previous works to have a solid set of technical indicators. This study aims to improve the prediction of whether the price per share will go up or down at the beginning of the next period by adding new features and leveraging a specific architecture to capture both long and short-horizon information in the stock market sequences by combining LSTM and GRU. Our work introduces a hybrid LSTM-GRU model, designed to capitalize on the strengths of both LSTM and GRU networks to handle extended sequences and simplify hyperparameter tuning. The hybrid model achieves more robust predictions by combining the LSTM's capacity for capturing long-range dependencies with the GRU's training efficiency. While this study concentrates on the LSTM-GRU approach, it acknowledges the significant progress made by transformer-based models like Transformer and BERT in natural language processing and time series forecasting through self-attention mechanisms. Research could further investigate incorporating these transformer-based models to improve stock market prediction accuracy.

Various combinations of indicators and requirement analyses are also discussed in the literature. Most of the works in the literature have not considered the impact of users' predictions or used it as a constant factor for all users. [3] proposes a robust stock market prediction model by integrating machine learning, sentiment analysis of news headlines, and fuzzy logic. Their methodology uses a machine learning module that applies Ridge Regression on historical stock data to predict closing prices. A Sentiment Analysis module collects and processes financial news headlines to determine

market sentiment. Finally, a Fuzzy Logic module combines the stock prediction and sentiment values to generate an investment recommendation. Research [4] is more detailed in classification and has used three strategies (1) (sell), 0 (hold), and -1 (buy) for classification. This study considered the use of deep learning models to extract features automatically. In both studies [3] and [4], the number of samples is not mentioned, making it somewhat more challenging to review the results. In [4], feature-based solutions, fuzzy clustering, and fuzzy neural networks were used for price prediction. This study approached the problem as a regression task to predict the US SSP 500 index price for 361 days. Moreover, the Root-mean-square deviation (RMSE) criterion was used. Table 1 shows the significant combinations of different features in the available research. It shows the difference between the proposed work and that of others.

In [8], the authors propose a hybrid autoregressive fractional integrated moving average (ARFIMA)-LSTM model to predict the abrupt stochastic variation of the financial market. The model extracts potential information from the residuals with the help of exogenous dependent variables and achieves better prediction accuracy by combining both models. Reference [9] proposed a classification method based on deep bidirectional LSTM-GRU for ciphertext classification, aiming to mark the category to which the ciphertext belongs. The weaknesses of the studied approaches can be summarized in the following areas:

- 1) Not considering social network information and only using market information increases the possibility of underfitting and overfitting the model [3], [4], [7].
- 2) The use of classification allows transactions that do not bring significant profits. For example, if the stock price of the next day is \$100 and today's price is \$99.80, it will be assigned the up, bullish, bull, or buy label [3], [4], [7].
- 3) Manual feature selection is time-consuming and challenging because the model becomes highly dependent on the selected features [6].

Research Highlights include:

- 1) Representing the features of Tehran stocks and using the features to improve prediction accuracy.
- 2) Representing the hybrid LSTM-GRU Model for improving the prediction accuracy.
- 3) Considering time series information of stocks for regression.

This study introduces a new hybrid LSTM-GRU model that leverages the advantages of both LSTM and GRU networks to handle long sequences and simplify hyperparameter tuning. This combined approach capitalizes on the LSTM's strength in capturing long-term dependencies and the GRU's training efficiency, leading to more robust predictions. The hybrid architecture simplifies hyperparameter tuning, a common deep learning challenge, as the GRU's simpler structure and fewer parameters facilitate this process. At the same time, the LSTM's ability to learn long-range dependencies

TABLE 1. A summary of recent studies on stock market prediction.

Reference	Data Type	Target Output	No. of Samples	Method	Performance measure
[5]	Dow Jones (Apple and Caterpillar)	market trend (bull/bear-flag)	20 years of daily data	Reinforcement learning + data mining techniques	Accuracy
[3]	Dow Jones	Trading strategy (1(sell),0 (hold),-1(buy))	-	feature selection +deep learning techniques	Accuracy
[4]	US SSP 500 index	stock price	361	feature selection+ fuzzy clustering +fuzzy NN	RMSE
[4]	Korea KOSPI200	market direction (up or down)	3650	feature selection+ ANN	statistical tests
[6]	US Dow Jones	market trend (bull/bear-flag)	91370	template matching	trading simulation
[7]	EUR/USD	sentiment analysis	25,793	text and image representation+ CapsuleNet	Accuracy, Precision, Recall, F1
The proposed study	Tehran Stock Exchange	Price prediction	8 stocks	LSTM+GRU	MAE, MSE, and RMSE

boosts predictive power. A rich feature space, incorporating technical analysis features like opening, closing, minimum, and maximum prices, captures market trends and fluctuations. Finally, potential future research directions, including integrating additional financial indicators and exploring transformer-based architectures, are discussed as potential avenues for improving model performance and real-world applicability. The rest of this paper is organized as follows. Section II presents related works on various stock market prediction methods. The employed datasets and their features and preprocessing steps are given in Section III. The proposed approach and related details are explained in Section IV. Section V discusses implementation and experimental results. Finally, the paper is concluded in Section VI, and ideas for future works are suggested.

II. RELATED WORK

Successful stock trading relies heavily on deciding to enter and exit in real time. This requires information review and expertise in exploring investment opportunities. As a result, there is an urgent need for an automated approach to effectively and efficiently use financial data to support investment decisions. In this section we review and discuss different approaches in predicting stock price.

A. MACHINE LEARNING BASED MODELS

Many researchers have employed computational and software techniques, including genetic algorithms and Artificial Neural Networks (ANNs) [10], [11], [12], [13]. These approaches successfully solve different problems and build many intelligent systems. One of their primary challenges is financial problems, as many researchers study the use of these techniques in finance [14].

Traditional technical analysis is based on some trading rules like high and low-price points in trends and is widely used in global markets. Unfortunately, stock trading systems that rely on technical rules are becoming less appropriate due to the dependence on traders' experiences for choosing

decision-making rules based on the purchase and sale of assets. Therefore, they cannot be used by all traders in the market. The main problem is that the technical rules are based on certain parameters. If these parameters are well-defined and intelligent, they will show accurate buy and sell signals for investor profitability. Another problem is that different technical rules may simultaneously generate conflicting trading signals, making decisions difficult. Refenes et al. [15] proposed modeling stock price behavior through neural networks and compared the performance with regression models. In this study, researchers used neural networks as an alternative to classical statistical techniques to predict stock prices in large corporations. This study indicated that neural networks outperform statistical techniques and create more successful models.

Tan et al. [16] designed a system for significant short-term stock price trend prediction. First, they carried out data preprocessing and then built a neural network model that calculated profitability status. In the study by Nauck and Kruse [17], they create a prediction system to buy, sell, or hold stocks in stock markets under an intelligent stock trading decision support system by using and integrating genetic algorithms based on fuzzy and ANNs. The developed system can identify qualitative interference variables in stock price prediction. Regardless of genetic algorithms, the researchers published an article with the same title in 1998. In this paper, researchers used a fuzzy Delphi (FDM) questionnaire to gather expert opinions on stock price prediction.

Yim [18] compared the econometric methods using ARMA and GARCH models. The results indicated the superiority of neural networks over the classic Autoregressive Moving Average (ARMA) and Generalized Autoregressive Conditional Heteroscedasticity (GARCH) models. The study of Abounouri et al. [19] compares the ANN and the Autoregressive Integrated Moving Average Model (ARIMA) in modeling and predicting the short-term exchange rate in Iran. Their results indicated that the used neural network has better predictive power than the ARIMA model. In another

study, Foroughi et al. [20] predicted earnings per share using a combination of ANNs and the Particle Swarm Optimization (PSO) algorithm. Their results indicated that univariate and multivariate models predicted each stock's profit with 78.5% and 91.7% accuracy, respectively.

Souto-Mairo et al. [21] used fuzzy logic to predict the direction of the Brazilian stock price index movement, and their prediction results were accurate. Researchers have employed artificial intelligence techniques such as genetic algorithms and ANNs for decades. Solving financial problems is one of the functional areas of artificial intelligence. Many researchers study the applications of artificial intelligence in finance. Alejandro Rodriguez et al. [22] used neural networks to enhance technical analysis indicators. Recent studies have used genetic algorithms for technical analysis to improve prediction parameters and strengthen Echo State Network (ENS) networks with improved parameters for predicting turning points. The results of studies [23] indicate that the approach used is better than the purchase and maintenance policy. Roodposhty et al. [24] apply the genetic algorithm for optimizing the portfolio of shares of investment funds on the TSE. The results indicated that the genetic algorithm could be more suitable for selecting developed portfolios. Aghamohammadi et al. [25] used neural network tools to check login duration time for stock trading with a technical approach. Their results indicated that the system's performance is suitable for bear markets. The proposed trading system and the maintenance and purchase method in the bull market were similar.

Michel et al. [26] used an embedded approach of Strongly Typed Genetic Programming (STGP) and Fuzzy Inference System (FIS) to generate trading rules. The authors used a FIS and STGP to generate trading rules for the US stock market, a FISTGP framework. According to the excess return with almost the same risk (1.28% more) for a 28.62% trial period, the results indicate that the proposed model is better than a buy-and-hold strategy.

Manahov [27] studied the effect of high-frequency data with the STGP approach to see if it benefits the market, finding that it could be better for corporate traders. Ha and Moon [28] adopted genetic programming (GP) to find patterns in the cryptocurrency market. Their study uses signals to increase the return on investment in the market. They indicated that GP could create applicable business rules and perform well in the experimental setup.

Davoodi et al. [29] researched predicting companies' stock prices on the Tehran Stock Exchange using an artificial intelligence algorithm (non-sensory parametric backup regression algorithm called Non-Paradox Non Parallel Support Vector Regression (PINSVR) in linear and nonlinear modes). The results indicate that the PINSVR algorithm in nonlinear mode has been able to predict stock prices more accurately over the years than in linear mode. Wang et al. [30] propose an interpretable intuitionistic fuzzy inference model (IIFI). While retaining prediction accuracy, the interpretable module in IIFI can automatically calculate the feature contribution

based on the intuitionistic fuzzy set. Results show that IIFI can effectively distinguish between important and redundant features by rating corresponding scores for each feature. In another study, Davoodi et al. [31] aimed to predict stock prices using the Chaid rule-based algorithm and particle swarm optimization (PSO). Due to the day-to-day growth of shareholders in the stock market and their attention to different stock prices in transactions, predicting stock price trends has become important. This research examined stock price prediction for the company (Chaid rule-based algorithm and PSO algorithm) from 2011 to 2016. The results indicate a significant relationship between earnings per share, E/P ratio, company size, inventory turnover ratio, and stock returns with stock prices. Additionally, the PSO algorithm has a good ability to predict stock prices. In this research, two new methods for solving this problem are proposed. Firstly, gold prices were used to predict and compare results with Goal variables instead of the traditional variable. Secondly, unlike previous studies, Deep Learning (DL) improves return prediction and compares results with average historical methods as a benchmark model. Diebold, Mariano, and West (DMW) statistics are used for statistical evaluation. The results indicate that the functional DP model has higher accuracy than the historical average model. It also indicates that improving out-of-sample predictions does not always depend on the number of the high input variables. On the other hand, this predictability can be improved using the price of gold as an input variable. The results also indicate that the price of gold is more accurate than the Goyal variable for out-of-sample return prediction.

B. USE OF DEEP LEARNING METHODS FOR FINANCIAL MARKET PREDICTION OF PRICE-BASED CLASSIFICATION MODELS

Karadaş et al. [32] present a multimodal stock price prediction approach combining historical financial metrics, tweets, and news articles. The study utilized a dataset that included three stocks representing different market states, gathering data from Yahoo Finance, Twitter, and The New York Times. The methodology involved sentiment analysis using ChatGPT-4o and FinBERT, notably incorporating a weighted sentiment score for tweets based on engagement. An LSTM model was used for prediction, integrating daily sentiment and article counts with price data. Results, evaluated by R^2 , Mean Absolute Error (MAE), and market simulation, showed that integrating these sources considerably increased forecast effectiveness, leading to higher returns compared to price data alone. Both sentiment models yielded similar results, with dominant technical indicators but sentiment features adding consistent value. The work highlights the substantial impact of sentiment analysis and offers a framework for multimodal financial forecasting. Wang [33] presents a novel contradictory-factor-based stability prediction model for the stock market, leveraging a sigmoid deep learning paradigm. The dataset for the case study included numerical stock

data for four organizations from Kaggle and information on seven influencing factors. The methodology involves mapping various influencing factors with profit outcomes and predicting stability using iteratively processed sigmoid and non-sigmoid deep learning layers. A key aspect is the dynamic linking of real-time trade data and adjusting the sigmoid function based on actual and predicted changes. The results, evaluated using metrics like prediction precision, stability matching, and RMSE against baseline models, show that the proposed model significantly outperforms methods like support vector machine (SVM), ANN, XGBoost, and LightGBM, achieving 96.12% prediction precision and a faster detection time of 0.05s. The work highlights a systematic approach to integrating diverse factors and deep learning for improved stock market stability forecasting, demonstrating high predictive power and reliability.

This paper [34] proposes a novel approach to enhance stock price prediction by leveraging generative adversarial networks (GANs) and transformer-based attention mechanisms. It utilizes a dataset of historical stock data from Yahoo Finance for Apple, Amazon, Google, and Microsoft and news sentiment data from Seeking Alpha. The methodology involves preprocessing incorporating FinBERT sentiment analysis and technical indicators, followed by a Variational Autoencoder (VAE) for denoising before feeding data into a GAN structure enhanced with transformer-based attention. Results, evaluated using RMSE and financial metrics, demonstrate commendable predictive capabilities compared to prior methods, effectively capturing market trends and showing applicability across the tested technology stocks. Key highlights include integrating multimodal data and advanced deep learning for more robust forecasting.

Gülmez et al. [35] propose a novel deep LSTM network optimized by the Artificial Rabbits Optimization (ARO) algorithm (LSTM-ARO) for stock price prediction. It uses a dataset of five years of Dow Jones Industrial Average (DJIA) index stock prices from Yahoo Finance, structured with 20 previous days predicting the next. The methodology involves ARO optimizing LSTM hyperparameters and architecture and comparison against models, including ANN and LSTM-GA. Results demonstrate that LSTM-ARO outperforms these models across key metrics like MSE, MAE, Mean absolute percentage error (MAPE), and R^2 , showing successful predictions. A highlight is the effectiveness of using ARO for optimizing deep learning models in stock market forecasting.

The authors in [36] have used a deep neural network to predict the price change over the next 5 minutes for 43 commodities and futures trading on Forex. Their input layer architecture has 9896 neurons for input features obtained from differences between the price drop and reciprocal movements of agreements. There are five fully connected layers for learning. The first hidden layer, in the four hidden layers, has 1000 neurons, and each subsequent layer has 100 neurons. They used backpropagation and stochastic

gradient descent (SGD) learning. Their results achieved an average accuracy of 42% for three-class classifications. They used progressive learning instead of recursive testing. They did not use transaction costs or the combination of supply and demand. All the predictions and features are based on the average price at the end of five minutes.

The study in [37] proposes three models combining Convolutional Neural Network (CNN) and RNN-based models for predicting the stock market index. The models were evaluated on the DAX, DOW, and SP500 indices. In contrast to existing studies, a novel input feature was used: the average of high and low prices. The performance of the proposed models was compared with conventional benchmark models regarding forecasting the closing prices of the stock market indices. The study of Zheng et al. [38] explores the effectiveness of machine learning techniques, specifically hybrid models, in capturing the nonlinear and dynamic nature of financial markets for predicting financial enterprise stocks and economic data trends. This study uses a novel hybrid model that combines a CNN, a Bidirectional Long Short-Term Memory network (BiLSTM), and an attention mechanism. Adding attention mechanisms is a key component of the model, which allows it to focus more on essential time steps or features during learning and prediction, improving accuracy and stability. The model's high accuracy is verified through comparisons with different algorithms, including LSTM-Attention, CNN-BiLSTM, and LSTM models, using MSE and MAPE as evaluation metrics. The improved hybrid model provides reliable technical support for financial applications such as quantitative trading. The CNN-BiLSTM-Attention model achieves an MSE of 0.013 and a MAPE of 0.0199, demonstrating higher reliability than previous models like LSTM-Attention, CNN-BiLSTM, and LSTM. They suggest that the new hybrid model CNN-BiLSTM-Attention model is better overall than the LSTM hybrid model. Reference [39] explores the potential of improving traditional stock market trading strategies by integrating them with LSTM neural networks. The research combines traditional trading strategies, such as MACD, Triple Exponential Moving Average (TEMA), Momentum (MOM), and Price Moving Average (PMA), with LSTM networks. LSTM networks are integrated into traditional strategies for price prediction, creating hybrid strategies. The performance of traditional strategies (without LSTM) is compared against the hybrid strategies (with LSTM) using backtesting on past data. A "buy and hold" strategy was used for additional comparison. Their work indicates that some trading strategies with integrated LSTM models performed better than traditional strategies, while others showed similar or slightly weaker results. The performance varies between different securities, suggesting that each stock reacts differently to specific trading signals within the trading strategies. The study of Zhao et al. [40] utilizes a hybrid CNN and LSTM model for stock price prediction. This study combines CNNs and LSTMs to capture short-term local features and long-term trends

in stock prices. The CNN is used to extract representative features from raw stock data. The LSTM then considers the impact of historical data on future prices. Data preprocessing, which includes cleaning, handling missing values, and feature engineering, is performed on stock data to improve the model's training and prediction performance. They collected a large dataset of historical trading data covering multiple periods and industries, including opening, closing, highest and lowest prices, and trading volumes. The proposed stock price prediction model provides valuable insights and new directions for research in stock market prediction.

The authors in [41] used a learned deep belief network (DBN) on technical indicators to trade the portfolio product of US Treasury Features. Their architecture has a DBN consisting of two stacked Restricted Boltzmann machine (RBMs). The first RBM is Gaussian-Bernoulli (15 nodes), and the second RBM is Bernoulli (20 nodes). DBN finds hidden features using different tripartite classifiers: Adjusted logistic regression, support vector machines, and a neural network with two hidden layers. If the product portfolio goes up within five days, they predict 1; otherwise, it equals -1. They trained the DBN using a proportional divergence algorithm. They calculate signals based on open, high, low, and close interest and volume data from 1985 to the 2008 financial crisis. They use 20 features: The daily trend is estimated and normalized over different periods. Applying Principal component analysis (PCA) in constructing the product portfolio is neutral towards the first principal component. The product portfolio is an artificial range of tools to trade that spreads between the ZF and ZN (5-year and 10-year US treasury note futures) contracts. All entry and moderate prices mean that the range of supply and demand is not considered. The results seem promising, with all three classifiers gaining 5-10% more accuracy than random predictions. The authors in [42] make trading decisions using the theory of the oscillation box based on DBN. The theory of the oscillation box states that stock prices will fluctuate within a specific range over some time. If the price moves outside this range, it enters a new box. The authors try to predict the ranges of this box. Their trading strategy involves stock purchase when the upper range is broken and stock sale when it falls below the lower range. Their architecture uses a DBN consisting of RBM and the rollback waiting in the final layer. Their learning uses blocked Gibbs sampling for greedy learning in each layer from the bottom up without an observer. Then, backpropagation is trained in a supervised manner to adjust the whole model. They selected 400 S&P 500 shares for the test, and the test suite covers 400 days from 2004 to 2005. They used open, high, low, and closed prices and technical analysis indicators for 14 model inputs. Some indicators show a more significant impact using Gray relational analysis or Gray correlation degree. According to their trading strategy, the results charged 0.5% of transaction costs per transaction and added parameters to stop the loss and adjust the transaction rate.

In [43], the effect of Twitter sentiment analysis on predicting trading volume, daily profit, and market fluctuations is examined. It also examines the traditional American Association of Individual Investors (AAII) indicators used for prediction and their relationship with Twitter sentiment analysis using various regression methods. It has been concluded that sentiment analysis in social networks can be a less expensive, faster, and more flexible alternative to traditional methods such as a referendum of stock market analysts and analyzing investors' opinions. The news texts in [44] are termed vectors and are used as input to the LSTM network. The vector transformation has been done and studied in several ways. The best results are obtained when the news headline is turned into an event and then into a vector using CNN. Additionally, market data has been turned into vectors. Finally, these inputs are fed into the LSTM network to provide the binary prediction of a decrease or increase of S&P 500 indices for the next day. The accuracy of this study is measured using confusion matrices, which calculate the ratio of correct predictions to the total evaluated data. The manual data labeling in the article [32] is used for teaching the SVM model. Comments on a Chinese stock exchange website were then emotionally labeled. These emotional and market data are used in the next step for the LSTM network to predict the reopening price of the China stock index. The best-reported accuracy is 88%, related to the reopening price prediction for the next day using SVM. Table 2 compares related works of stock prediction and our approach.

Shaban et al. [45] proposed a stock market prediction approach based on deep learning (SMP-DL), which divides the whole process into two stages of preprocessing and price prediction. They utilized bots to scrape data from online resources and store it in the required formats. Then, the data is cleaned, ensuring no missing values remain in the samples. In the next stage, min-max normalization is performed to scale the data to a specific range. The proposed LSTM-BiGRU model could outperform baselines in predicting the closing price of the stock market. A mobile application is also proposed for real-time trading that allows investors instant access to market data and useful insights and analyses. In a work by Sivadasan et al. [46] they were able to use a hybrid RNN model to predict the fluctuation of energy demand forecasting. The energy demand time series tends to be more stable than the stock time series. In another work, Li and his colleagues [47] have used a composition of time series forecast to make it simple for the forecast model to predict time-series fluctuations. However, they had to use different techniques to reduce the computational shortcomings of their proposed model, resulting in many hyperparameters. Reference [48] leveraged a modal decomposition technology, Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN), to decompose original stock price data into multiple intrinsic mode functions (IMFs) to reduce noise and error. This hybrid (CEEMDAN-LSTM-SA-LightGBM) model uses an LSTM

network to predict IMFs individually. Although the accuracy of the LSTM model increased, it also exerted computational costs. They utilized a Light Gradient Boosting Machine (LightGBM) to improve prediction speed, efficiency, and stability. A Simulated Annealing (SA) algorithm is incorporated in this approach to optimize hyperparameters in LightGBM, which can significantly reduce computational time. Lu et al. proposed a Time-series Recurrent Neural Network (TRNN) [49] focusing on time, price, and the volume of trades. Trading volume is considered a cardinal determiner of price variations in this study. They found that using data compression can make the learning process more effective, while not using any compression obstructed the process. Additionally, a sliding window algorithm is employed to increase prediction accuracy and extract trends and turning points. The reason for using RNN as the backbone of the model is to take all the sequences into account and, at the same time, consider the recent volumes more effective in determining current prices. Ensemble Empirical Mode Decomposition (EEMD)-ensemble CNN by Das et al. [50] consists of four main components. X (formerly Twitter) sentiment analysis, ensemble CNN, EEMD, and technical indicators are combined to improve the accuracy of stock price prediction further. They gathered posts that included specific equities and cleaned them only to contain important information suitable for sentiment analysis. These posts can exhibit valuable information about public opinion toward the market for future price prediction. A CNN model is used to classify each post as positive, neutral, or negative. Technical indicators, such as mathematical calculations on various features of historical market data, are used to provide a more in-depth understanding of the current market status. RSI is used in this study to determine the rate of price changes and movements, which will be discussed in Section III-A4. EEMD is used to extract IMFs to represent the underlying patterns in the time-series data. An ensemble CNN model consisting of sub-networks is used to process the IMFs and learn various training data features. MVL-SVM by Long et al. [51] uses a multi-view learning approach combined with SVM to enhance stock price prediction. They used news in addition to market data and leveraged a multi-kernel learning framework to build their model. They gathered unstructured Chinese news and then preprocessed it before using it for training. In the preprocessing step, they removed noise and redundant information, such as stop words or punctuation, and cleaned the data for each business day. MVL-SVM can fuse the two different data sources without being impacted by the inconsistencies of the data. They applied their method across various periods using sliding windows ranging from one to five days. They concluded that only considering the news for a one-day can significantly improve performance. Their findings indicated that their proposed approach yields superior performance compared to baselines.

The hybrid LSTM-GRU architecture, designed to leverage the strengths of both GRU and LSTM networks for enhanced

predictive power, introduces inherent complexity that can result in significant computational demands. Although GRU layers contribute to computational efficiency by handling short- and medium-term dependencies at the initial stage, the combined model possesses approximately 1.5 times the learnable parameters compared to individual LSTM or GRU networks. This increased complexity necessitates substantial processing power, memory, and training time, which can pose practical challenges, particularly concerning accessibility for resource-constrained organizations. While a detailed empirical runtime analysis across different hardware configurations or an extensive discussion on scalability beyond the need for architectural optimization and parallel processing for large datasets were not within the primary scope of the current study, we acknowledge their importance for real-world deployment. Addressing these practical implementation challenges, including computational cost and scalability, is recognized as a crucial direction for future research to improve the model's efficiency and applicability.

III. DATASETS

We collected data for eight different symbols. Table 3 shows the data details and their collection intervals. This data was collected from 2012 to 2024. In addition to the main features (open, high, low, close, volume), 42 features were calculated during data preprocessing. These features serve as inputs to the models, along with the main features, which are shown in Table 4. The 42 technical indicators have been classified into several primary categories to provide a structured overview based on their core functions in analyzing market data. These categories include:

- **Trend Following Indicators:** aim to identify the direction of price trends,
- **Momentum Oscillators:** measure the speed and rate of change of price movements,
- **Volatility Indicators:** assess the degree of price fluctuation,
- **Volume Indicators:** analyze the strength of price movements based on trading volume,
- **Others:** a category for those indicators that exhibit hybrid characteristics or do not fit neatly into the mentioned groups.

In the following, the features of the dataset and the preprocessing step are discussed.

A. FEATURES

Technical indicators are very common in predicting stock market movement, so 42 indicators enhance our prediction. The technical indicators are calculated from the data of eight stocks using the Python library Stockstats. The most common indicators used are as follows:

1) SIMPLE MOVING AVERAGE (SMA)

An SMA is exactly what it sounds like. It calculates the day-to-day average, or mean, of a stock's price over time. It is

TABLE 2. The presented papers based on the combination of socio-emotional economic analysis of the stock market.

Author and year of publication	Predicted variable	Input data	Solution used	Index and accuracy reported
Oliveira (2017)[43]	Daily profit, trading volume, market fluctuation, AAI, and II Indicators	Emotion analysis, market data	Types of machine learning methods(SVM,NN,RF,NN)	NMAE=8
Vargas (2017) [44]	S&P500 indicator	News texts and market data	Word2vec, Sentence2vec,CNN, LSTM	Accuracy=65%
Li(2017)[36]	CSI300 indicator	Emotion analysis of the posts registered in the sites for exchanging stock information and market data	Naïve Bayes, LSTM	Accuracy=87%
Chong (2017) [52]	Move price per share up and down	Price data per share once every 5 minutes	Data representation, DNN	NMSE=0.96 MAE=0.5
Chakraborty (2017) [53]	The rate of going up or down in Apple stock prices and DJIA indicator	Recorded tweets about Apple and the total stock index	SVM, boosted regression tree	RMSE(AAPL)=1.2 RMSE(DJIA)=82
Kamkarhaghighi (2016) [54]	The connection between the validness of a Twitter user and user followers information	Profile information of some active users in the stock exchange field and daily stock prices of several companies	Calculate the prediction of each user's opinion and Naïve bayse	F-Measure =0.66
Sohangir (2018) [55]	dictionary-based comparison of machine learning methods in stock market prediction	sentiment analysis stock twits	SVM Naive Bayes Sentiwordnet	Accuracy=87%
Oliveira (2016) [56]	Twitter sentiment analysis	Registered Tweets with polarity tags in stocktwits	TF-IDF, IG, PMI	Accuracy=73%
Kedar (2021) [57]	Twitter sentiment analysis for stock prediction	Registered Tweets with polarity tags in StockTwits	Arima	Accuracy=79%
Carvalho (2021) [58]	Tweet sentiment analysis for stock prediction	Tweets including joint information of corona and stock	SenticNet	MAE=0.55%
Huang (2021) [59]	Twitter sentiment analysis for stock prediction	Features related to stocks and tweets	Use the LSTM model for stock prediction	Recall=0.92%
Efrilianda(2021)[60]	Twitter sentiment analysis for stock prediction	Tweet	Fuzzy model	-

TABLE 3. Collected data for each stock and their collection interval.

Symbol name	Industries Group	Date of the first collected data	Date of the last collected data	Total collected data	day counts of symbols stop
D Zahravi (zahravi pharmaceutical company)	pharmaceutical	March 21, 2012	March 20, 2024	1117 days	98 days
D Tolid (Tolid Darou pharmaceutical company)	pharmaceutical	March 21, 2012	March 20, 2024	1117 days	68 days
V BMLT (MELLAT BANK)	investment	March 21, 2012	March 20, 2024	1117 days	168 days
Atlas (investment funds of Mofid securities)	investment	March 21, 2012	March 20, 2024	1117 days	61 days
khodro (Iran Khodro)	automobile manufacturing	March 21, 2012	March 20, 2024	1117 days	78 days
Kha Motor (Iran tractor manufacturing Motor-sazan company)	automobile manufacturing	March 21, 2012	March 20, 2024	1117 days	42 days
Qazvin (Qazvin Sugar Factories Company)	Food stuffs	March 21, 2012	March 20, 2024	1117 days	61 days
Gh Minoo(East Minoo Food Industrial Co)	Food stuffs	March 21, 2012	March 20, 2024	1117 days	41 days

shown in Eq. (1).

$$SMA_g = \frac{C_1 + C_2 + \dots + C_n}{n} \quad (1)$$

2) MOVING AVERAGE INDICATOR (MA)

It is one of the most popular technical indicators used to identify market price trends.

3) MOVING AVERAGE CONVERGENCE DIVERGENCE (MACD)

The MACD is a technical momentum oscillator plots two Exponential Moving Averages (EMAs). One is subtracted from the other to create a signal line or “divergence” (MACD Line) and then added back to it (signal). Three main parameters are signal length, moving average convergence/divergences frequency, and periodicity. By default,

TABLE 4. Technical indicators and their functions.

Category	Indicator	Primary Function/Contribution to Price Prediction
Trend Following	Simple Moving Average (SMA)	Smooths price data to identify trend direction based on average past prices.
	Exponential Moving Average (EMA)	Similar to SMA but weights recent prices more heavily for quicker trend identification.
	Volume Weighted Moving Average (VWMA)	Averages price weighted by volume, indicating trend direction with consideration for trading activity.
	Linear Regression Moving Average (LRMA)	Uses linear regression to fit a trend line to prices, potentially offering a more accurate trend representation.
	Kaufman's Adaptive Moving Average (KAMA)	Adjusts its smoothing based on market volatility to better identify trends in varying market conditions.
	Ichimoku Cloud	A comprehensive system identifying trend, support, resistance, and momentum using multiple lines and a cloud.
	Elder-Ray Index (ERI)	Measures buying and selling power relative to an EMA to assess trend strength.
	Correlation Trend Indicator (CTI)	Likely uses correlation to assess the strength and consistency of a trend.
Momentum Oscillators	Raw Stochastic Value (RSV)	Measures the position of the current price within its recent high-low range.
	Relative Strength Index (RSI)	Measures the speed and change of price movements to identify overbought/oversold conditions.
	Stochastic Oscillator (KDJ)	A stochastic indicator with an added J line for increased sensitivity in identifying short-term momentum.
	Moving Average Convergence Divergence (MACD)	Shows the relationship between two EMAs to identify momentum shifts and potential trend changes.
	Williams Overbought/Oversold index (WR)	Measures overbought/oversold levels on an inverted scale, similar to Stochastics.
	Commodity Channel Index (CCI)	Identifies overbought/oversold levels based on price deviation from its average over a period.
	Triple Exponential Moving Average (TRIX)	Measures the percentage change in a triple-smoothed EMA to identify momentum shifts.
	Percentage Price Oscillator (PPO)	Measures the difference between two EMAs as a percentage of the longer EMA, normalizing momentum by price.
	Stochastic RSI (StochRSI)	Applies the stochastic oscillator formula to RSI values for increased sensitivity.
	LazyBear's Wave Trend (WT)	Likely identifies trend direction and potential wave-like price movements using smoothed data.
	Aroon Oscillator	Measures the time since the last high and low to assess trend strength and potential reversals.
	Awesome Oscillator (AO)	Measures momentum by comparing short-term and long-term SMAs smoothed by median price.
	Rate of Change (ROC)	Measures the percentage change in price over a specific period, indicating the speed of price movement.
	Coppock Curve	A weighted sum of ROC used to identify long-term buying opportunities.
	Relative Vigor Index (RVGI)	Compares closing price to opening price within the trading range to assess the vigor of price movement.
	Know Sure Thing (KST)	Combines smoothed ROC indicators over multiple timeframes for a comprehensive momentum signal.
	Pretty Good Oscillator (PGO)	Measures the percentage difference between two moving averages to identify momentum shifts.
	Psychological Line (PSL)	Measures the ratio of rising periods to total periods, reflecting market sentiment and potential overbought/oversold conditions.
	Percentage Volume Oscillator (PVO)	Measures the percentage difference between two volume moving averages to identify changes in volume momentum.
Volatility Indicators	Moving Standard Deviation (MSTD)	Measures the dispersion of price data around its moving average.
	Moving Variance (MVAR)	Measures the squared deviation of price from its moving average.
	Bollinger Band (Bolling)	Plots bands based on standard deviation around a moving average to indicate price volatility.
	True Range (TR)	Measures the greatest of several price differences to capture the full daily price range, including gaps.
	Average True Range (ATR)	Averages the True Range over a period to measure overall market volatility.
	Choppiness Index (CHOP)	Measures whether the market is trending or moving sideways.
	Kaufman's efficiency ratio (KER)	Quantifies market noise by measuring the efficiency of price movement.
Volume Indicators	Volume Variation Index (VR)	Compares volume on up days to volume on down days to gauge buying and selling pressure.
	Money Flow Index (MFI)	Measures the flow of money into and out of an asset using both price and volume.
	Balance of Power (BOP)	Measures the strength of buyers versus sellers within a trading day based on the closing price's position in the range.
Other Indicators	Difference of Moving Average (DMA)	Measures the difference between moving averages of different periods or a displaced moving average to analyze trend and momentum.
	Directional Moving Index (DMI)	A system of indicators (+DMI, -DMI, ADX, ADXR) used to assess trend direction and strength.
	Smoothed Moving Average of ADX (ADXR)	A smoothed version of ADX that measures the change in trend strength.
	Mean Absolute Deviation (MAD)	Measures the average absolute difference from the mean, providing a robust measure of volatility.

these values are 12, 26, and 9, respectively. The formula to calculate MACD is as Eq. (2):

$$\begin{aligned} MACD &= EMA_{12}(c) - EMA_{26}(c) \\ SignalLine &= EMA_9(MACD) \end{aligned} \quad (2)$$

4) RELATIVE STRENGTH INDEX (RSI)

It is a technical momentum indicator that compares the magnitude of recent gains and losses over time and plots them as an oscillator. The RSI was developed in 1978 and has become one of the most popular oscillator indicators. It is shown in Eq. (3).

$$\begin{aligned} RSI &= 100 - \frac{100}{1 + RS} \\ RS &= \frac{\text{Average Gain}}{\text{Average Loss}} \end{aligned} \quad (3)$$

5) BIAS

The formula to calculate BIAS is in Eq. (4):

$$BIAS(n) = \frac{C - MA_n(C)}{MA_n(C)} * 100 \quad (4)$$

where C is the Current closing price, and $MA_n(C)$ is the moving average of the n -day closing price. BIAS is a technical indicator that reflects the deviation between the closing price and MA by calculating the percentage difference in a certain period. It indicates the possibility of price retracting or rebounding due to deviating from the moving average trend in severe volatility and the credibility of price moving within the normal range of volatility to continue the original trend. In this paper, it is set to 5.

6) ON BALANCE VOLUME (OBV)

The formula to calculate OBV is in Eq. (5):

$$OBV(t) = \begin{cases} OBV(t-1) + vol(t) & \text{if } c(t) > c(t-1) \\ OBV(t-1) - vol(t) & \text{if } c(t) < c(t-1) \\ OBV(t-1) & \text{if } c(t) = c(t-1) \end{cases} \quad (5)$$

where $OBV(t)$ is OBV at time t , $Vol(t)$ is Trading Volume at time t , and $C(t)$ is Closing Price at time t . OBV is a technical index that takes volume as a breakthrough to find hot stocks and analyze stock price movement trends.

7) VOLATILITY RATIO (VR)

The formula to calculate VR is in Eq. (6).

$$\begin{aligned} VR &= \frac{AVS + 0.5 * CVS}{BVS + 0.5 * CVS} \\ AVS &= \sum_{n=1}^{10} AV_n \quad BVS = \sum_{g=1}^q BV_n \quad CVS = \sum_{n=1}^r CV_n \end{aligned} \quad (6)$$

where $p + q + r = N$, AV_n is the volume when the price rises during N days, BV_n is the volume when the price falls

during N days, and CV_n is the volume when the price remains unchanged during N days. VR is a strength index that mainly measures the stock price trend from the volume perspective.

8) OTHER FEATURES

In addition, some other features were by default in our datasets prepared from the Iran stock market. Some of these include:

- Number of trades: This element shows the number of transactions done on the last day of a share.
- Volume of trade: This element shows the whole value of trades on the last day of a share.
- Close price: This element shows the price of a share traded for the last time on the last day.
- Minimum and Maximum price: These two elements show the minimum and maximum price for a share that has been traded.
- price first: This element shows the first price of a share after the market starts.

Other features include the last price and date.

B. PREPROCESSING

Initially, the datasets have some invalid data. This was due to holidays and days when trading for some symbols was stopped. So, we used the null data removal method. Then, feature selection is applied to the data as another preprocessing step. For this purpose, Random Forest (RF), an embedded method for selecting features, is used. The primary objective was to reduce dimensionality by identifying the most relevant indicators and observing the results with a more limited feature set. Our analysis indicated that models trained on RF-selected features could achieve almost the same results as those trained on the full feature set of 42 technical indicators, suggesting that while feature selection is viable, the comprehensive set provides valuable information captured by the model. It is worth noting that while Random Forest provided insights into feature importance for selection, a comparative evaluation of this feature selection approach against other distinct dimensionality reduction methods, such as Principal Component Analysis (PCA), was not performed in this study. RF was generally used in the preprocessing stage to reduce dimensionality and to observe the results with a more limited feature set. We observed that models trained on RF-selected features could achieve almost the same results as those trained on the full feature set. The data are not scaled, so Min-Max normalization was used to normalize them.

IV. METHODOLOGY

RNNs and deep learning-based methods have also been proposed for stock price prediction. It is better than other methods among the results taken from LSTM-based methods. LSTM is an improved version of recurrent networks. However, the GRU algorithm has recently been proposed to improve these networks. In this paper, the accuracy of fluctuation prediction is improved by combining LSTM and

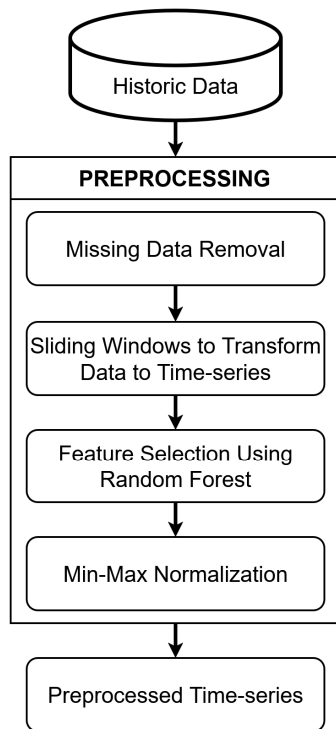


FIGURE 1. Preprocessing pipeline.

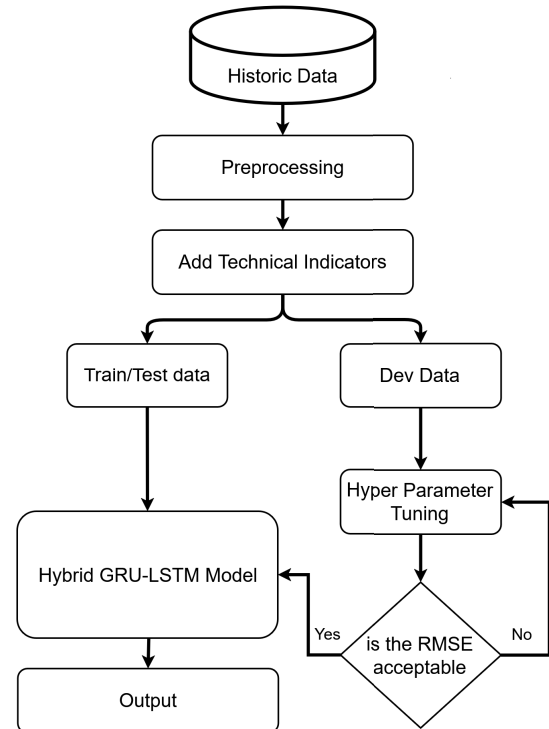


FIGURE 2. Proposed structure for stock trend prediction.

GRU networks due to the excellent performance of GRU networks. In the combination process, the output of the GRU layer is sent to the LSTM layer, and the output of the LSTM layer reaches the fully connected layers after the combination. GRUs are computationally efficient and effective in capturing short- and medium-term temporal dependencies. By placing GRU layers at the beginning of the architecture, we reduce the computational overhead while efficiently extracting the most relevant patterns from the input data. GRUs also help mitigate the vanishing gradient problem, enabling better gradient flow for initial feature learning. LSTM architecture, on the other hand, is particularly well-suited for capturing long-term dependencies due to its more complex gating mechanisms, which allow for finer control over the flow of information. By placing LSTM layers after the GRU layers, we can refine the extracted features and focus on long-term trends and dependencies that are vital for stock prediction tasks where it's influenced by short-term patterns (e.g., market fluctuations) and long-term trends (e.g., seasonal or economic cycles). The proposed hybrid architecture enables us to model this dual nature of stock price behavior effectively. The output of fully connected layers is also sent to the regression layer to perform the prediction process. Fig. 2 illustrates the generalities of the proposed method.

The proposed method is a four-layer structure. These four layers at a glance are:

- 1) Input layer: Each input feature related to the prediction is given to GRU inputs in this layer.

- 2) GRU layer: The inputs of the GRU layer are vectors taken from the input layer. The third proposed approach has no LSTM layer, so it is sent directly from this layer to the output layer.
- 3) LSTM layer: The inputs of the LSTM layer are vectors taken from the GRU layer. The second proposed approach has no GRU layer, so it is sent directly from this layer to the output layer. In this layer, the connection between the previous and current layers is made through gates. Logically, the inputs of both LSTM and GRU models are equal. To connect LSTM layers to GRU, it is enough to set the value of the return sequence to true.
- 4) Output layer: At first, the output of the LSTM network is flattened. The data normalization operations of the proposed models will be performed after receiving the research dataset. Then, the model learning process begins using the GRU-LSTM hybrid network. This step considers A GRU block for each feature of the research and learns feature weight and feature weight toward the previous feature simultaneously to predict fluctuations in the stock market. Then, the output of the GRU layer reaches the LSTM layer, and LSTM sends the result to the fully connected layer after input processing. The proposed hybrid LSTM-GRU model employs a sequential architecture where GRU layers precede LSTM layers. The rationale for this design is rooted in the distinct capabilities of these networks. GRUs are known for their computational efficiency

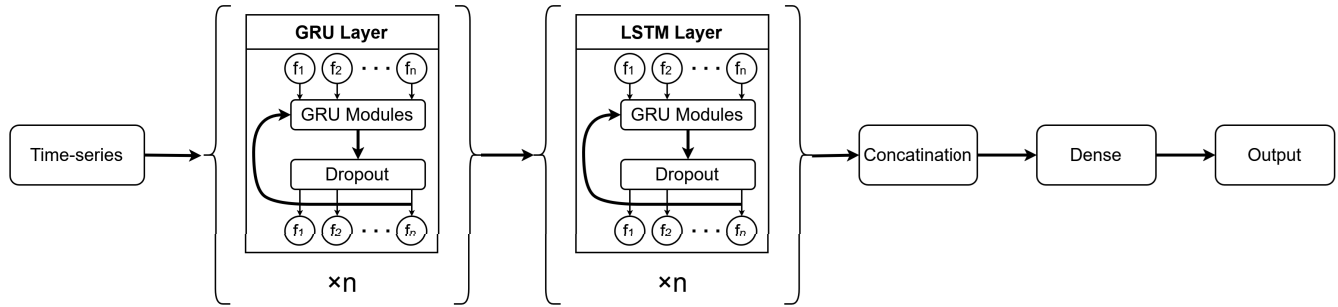


FIGURE 3. Hybrid LSTM-GRU model architecture.

and effectiveness in capturing short- and medium-term temporal dependencies. By positioning GRU layers at the initial stage, the model can efficiently extract relevant patterns from the input data while mitigating vanishing gradients. The output of the GRU layers is then fed into the LSTM layers. LSTM networks are particularly adept at capturing long-term dependencies due to their more complex gating mechanisms. This sequential arrangement allows the LSTM layers to refine the features learned by the GRUs, focusing on the long-term trends and dependencies crucial for forecasting stock prices, which are influenced by short-term fluctuations and long-term market cycles. While other architectures, such as bidirectional or parallel arrangements of GRU and LSTM or models incorporating attention mechanisms, exist and offer potential benefits, these were not explored in the design of the current hybrid model. However, these alternative architectures and the integration of attention mechanisms are considered promising directions for future research. The pseudocode of the proposed approach is presented in Algorithm 1.

The parameter space of deep learning methods is entirely continuous. This means the model performance changes drastically depending on the selected parameters in the continuous space. For considering parameter optimization for learning rate, number of layers, and sliding window length, we have used a dev set in our experiments for 10% of generated time series from historical data. For robust train and test split, we have used 70 percent of first data from 2012 to 2024 for the train set and the rest 20 percent for the test set and 10 percent for the dev set. Therefore, we used sliding windows to get n ($n=10$) previous data points to predict the target data for training. We have used the same approach for the test set, so we have used n previous data points to predict the target data point, but the model has not seen the last 25 percent of the data interval from 2012 to 2024 during the training phase. Here are some of these parameters:

- 1) Number of layers: The number of layers in a deep network determines its depth. The layers of a convolutional network are categorized as convolution, pooling, and fully connected, and LSTM, RNN, and

Algorithm 1 The Proposed Approach

```

Data  $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)\}$ 
 $D_{cleaned} \leftarrow$  missing data removal( $D$ )
 $S \leftarrow$  sliding windows size
Preprocess:
 $Data_{proc} \leftarrow Slidewindow(D_{cleaned})$ 
 $Time - series D_t =$ 
 $\{(x_1, x_2, \dots, y_1), \dots, (x_{m-s}, x_{m-s-1}, \dots, y_m)\}$ 
 $F_s \leftarrow$  subset of features using random forest
 $D_{norm} \leftarrow$  minmax normalize( $D_t$ )
 $D_f \leftarrow$  add technical indicators( $D_{norm}$ )
Process:
 $L_1 \leftarrow$  number of GRU modules
 $L_2 \leftarrow$  number of LSTM modules
 $T \leftarrow$  Number of learning rounds
 $X_{train}, X_{test}, X_{dev} \leftarrow DataSplit(D_f, 0.70, 0.20, 0.10)$ 
For  $linT$ 
 $\{GRU_{module\_input}\} \times L_1 \leftarrow input\_features$ 
 $\{LSTM_{module\_input}\} \times L_2 \leftarrow GRU_{modules\_output}$ 
 $flatten_{output} \leftarrow flatten(LSTM_{modules\_input})$ 
 $concatenated_{output} \leftarrow concatenator(flatten_{output})$ 
 $regression \leftarrow sigmoid(concatenated_{output})$ 
EndFor

```

GRU specify the number of layers in a recurrent network. Increasing the number of layers can sometimes improve model performance.

- 2) Learning rate: The learning rate in optimization indicates the step size in minimizing the loss function.
- 3) Number of neurons in the fully connected layer: This value can also change depending on the problem and achieve different accuracies.
- 4) The n units of GRU and LSTM are considered ($n=50$) in experiments. the output of each unit in a layer is the input of the corresponding unit in the next output after applying the layer dropout.

The steps in the proposed approach are as follows:

- (a) Receiving the research dataset: The examined data in the data collection section is considered model input.

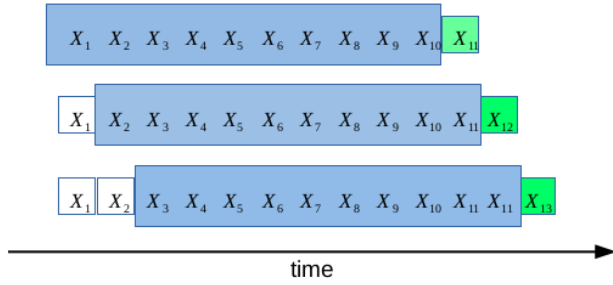


FIGURE 4. The idea of sliding window algorithm.

(b) Initially, there are invalid data in the datasets. This was due to holidays and days when trading for some symbols was stopped. So, we used the null data removal method.

(c) Formation of sliding windows: The formation of sliding windows for predicting stock market fluctuations is shown in Fig. 4.

(d) Data normalization: Normalization is necessary for the data related to the features, considering the collected datasets do not have the same type and domain. Min-max normalization was used for this purpose. To mitigate the negative impact of outliers and noise on normalization and model training, we employed a straightforward technique: data points falling outside the interquartile range (between the 25th and 75th percentiles) were treated as outliers and excluded from the analysis.

Fig. 1 illustrates the full preprocessing stage of the proposed method. (e) Adjusting parameters: GRU and LSTM networks usually have three primary parameters: the number of epochs (complete passes of the dataset during the training phase), the number of neurons in the neural network, and window length (w). To prevent overfitting, we have used dropout layers after every layer in the model. In each module. Fig. 3 illustrates the full Hybrid LSTM-GRU model architecture. These parameters are selected by optimizing the price prediction of the desired currencies.

To optimize the model's hyperparameters, several experiments were conducted. Epoch values between 50 and 200 were tested, with the optimal value determined based on validation set performance. The number of neurons in the GRU and LSTM layers was iteratively increased from an initial value of 32 until performance gains plateaued or overfitting occurred. The Adam optimizer was used to train the model, with learning rates of 0.001, 0.005, and 0.01 explored. The sliding window length, which governs the amount of historical data used for prediction, was also optimized. Starting with shorter windows, the length was gradually increased, balancing the need to capture long-term trends with sensitivity to recent fluctuations. The chosen window length maximized validation set performance while considering the computational cost associated with larger windows. Each layer contained 128 units, a value determined through experimentation and prior research as balancing performance and computational cost; further increases yielded negligible improvement. Overfitting was mitigated using a

0.2 dropout rate, a common and effective regularization technique. The Adam optimizer employed a 0.001 learning rate, a standard value balancing convergence and stability. A batch size 64, also chosen through experimentation and prior work, provided a good compromise between training speed and model performance. Finally, a 30-day sequence length was used for the input data, capturing sufficient historical information without excessive computational demands. The inclusion of a comprehensive set of technical indicators is motivated by the need to capture diverse market dynamics, especially in less-studied markets like the TSE, where a solid standard set of indicators is not universally established. We explored feature selection using Random Forest during preprocessing and found that models trained on the reduced feature set achieved performance comparable to those using the full set. The empirical evaluation of the impact of features, as illustrated for the Atlas stock, focused on sets of features (e.g., Momentum, Volatility) rather than individual indicators, demonstrating that utilizing the complete set of features generally yielded the best results for this dataset. (e) Splitting the research datasets into training and test sets: Data volume was considered 70% for training, 20% for testing, and 10% for validation.

(f) Send input to the GRU layer: The inputs of the GRU layer are vectors taken from the input layer. If these vectors are displayed as $Out_{embed} = [X_1, X_2, \dots, X_n]$, these vectors are calculated using Eq. (7).

$$\begin{aligned} f_t &= \sigma(W_f x_t + U_f h_{t-1} + b_f) \\ r_t &= \sigma(W_r x_t + U_r h_{t-1} + b_r) \\ \tilde{C}_t &= \tanh(W_C x_t + U_C (r_t \odot h_{t-1}) + b_C) \\ h_t &= f_t \odot h_{t-1} + (1 - f_t) \odot \tilde{C}_t \end{aligned} \quad (7)$$

x_t represents the input features at time step t , h_t is the hidden state at time step t , representing the memory of the GRU, z_t is the update gate, which determines how much the hidden state should be updated, r_t is the reset gate, which determines how much of the past hidden state to forget, h'_t is the candidate hidden state, σ is the sigmoid activation function, w_z , w_r , and w_h are the weights for the input x_t , U_z , U_r , and U_h are the weights for the hidden state h_{t-1} , b_z , b_r , and b_h are the bias terms, and $\odot$ represents element-wise multiplication.

(g) Send the output of the GRU layer to LSTM: The LSTM input in the T step is the vector $X \in R$. It is calculated from Eq. (8).

$$\begin{aligned} f_t &= \sigma(W_f x_t + U_f h_{t-1} + b_f) \\ i_t &= \sigma(W_i x_t + U_i h_{t-1} + b_i) \\ \tilde{C}_t &= \tanh(W_C x_t + U_C h_{t-1} + b_C) \\ C_t &= f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \\ o_t &= \sigma(W_o x_t + U_o h_{t-1} + b_o) \\ h_t &= o_t \odot \tanh(C_t) \end{aligned} \quad (8)$$

where x_t represents the input vector from the GRU layer at time step t , h_t is the hidden state at time step t , i_t is the input

TABLE 5. List of symbols used in Eq. (7) and Eq. (8).

Symbol	Description
x_t	Input at time step t
h_t	Hidden state at time step t
C_t	Cell state (only in LSTM)
$\tilde{C}_t$	Candidate state (used in both LSTM and GRU)
f_t	Forget gate in LSTM / Update gate in GRU
i_t	Input gate (only in LSTM)
o_t	Output gate (only in LSTM)
r_t	Reset gate (only in GRU)
W_f, W_i, W_o, W_r, W_C	Weight matrices
U_f, U_i, U_o, U_r, U_C	Recurrent weight matrices
b_f, b_i, b_o, b_r, b_C	Bias terms
$\sigma(\cdot)$	Sigmoid activation function
$\tanh(\cdot)$	Hyperbolic tangent activation function
$\odot$	Element-wise multiplication

gate, which controls the flow of new information into the cell state, f_t is the forget gate, which controls the amount of past information to discard from the cell state, g_t is the cell state update, o_t is the output gate, which controls the amount of information to output from the cell state, σ is the sigmoid activation function, W_i , W_f , W_g , and W_o are the weights for the input x_t and hidden state h_{t-1} , b_i , b_f , b_g , and b_o are the bias terms, $\tanh$ is the hyperbolic tangent activation function.

(h) Send output to fully connected layer: In the final step, the output is sent to the fully connected layer.

V. IMPLEMENTATION AND RESULTS

The performance of the proposed method is evaluated based on the evaluation criteria in this section after introducing the implementation environment and data preprocessing. To further enhance the robustness of our results, we conducted k-fold cross-validation (k=5) for both our model and the baseline models. This technique helps to assess the model's generalization performance. It reduces the risk of overfitting. We have conducted 10 experiments for each stock and averaged them for unbiased results. We report the average performance metrics (MSE and MAE) and their standard deviations across the k folds for each model. By incorporating k-fold cross-validation, we provide more compelling evidence for the superiority of our proposed hybrid LSTM-GRU model and enhance the credibility of our findings. Keras [61] has been used for developing the described models. Keras is a high-level library for building neural network tools written in Python and can run on TensorFlow, CNTK, and Theano. Keras includes several implementations of neural network structure blocks such as layers, objects, activation functions, optimizers, and numerous tools for working with images and text data. The data collected from Iranian stocks includes historical information, for which several indicators were used. Different MSE, RMSE, and MAPE approaches on different stocks are shown from Fig. 7 to 33. The proposed LSTM approach, compared to MSE, reduced the error into a logarithmic form. The GRU approach has often led to overfitting, which is less common in hybrid LSTM-GRU. In the Kh Motor stock, the results obtained by LSTM are better than those of the proposed approach. This is

TABLE 6. CNN approach characteristics reported in [56].

Layer	Output size
Conv1d	(None, 4, 32)
Conv1d	(None, 4, 64)
Leaky ReLu	(None, 4, 64)
Max pooling	(None, 2, 64)
Conv1d	(None, 2, 128)
Leaky ReLu	(None, 4, 128)
Max pooling	(None, 1, 128)
Flatten	(None, 128)
Dense	(None, 256)
Leaky ReLu	(None, 256)
Dropout	(None, 256)
Dense	(None, 1)

due to the high volatility of this share and the lack of sufficient data for the hybrid network. The hybrid network has about 1.5 times the learnable parameters of the LSTM and GRU networks. Moreover, in the absence of sufficient data, these learning parameters cause the model to depend too much on the data and show sensitivity; this usually causes overfitting. Implementing the hybrid LSTM-GRU model for stock market prediction faces practical challenges related to computational resources, scalability, and interpretability. The model's complexity demands substantial processing power, memory, and training time, potentially limiting accessibility for organizations with limited hardware. Scaling the model to handle ever-increasing datasets requires architectural optimization and parallel processing techniques. Maintaining performance with continuous data ingestion necessitates regular updates and retraining. Furthermore, the model's complex nature makes hyperparameter tuning difficult and hinders interpretability, creating a "black box" effect that can be problematic for stakeholders requiring explainable predictions. Future research should focus on more efficient architectures and techniques to overcome these limitations and improve the model's real-world applicability.

The results are gathered regarding evaluation criteria on different datasets. The proposed hybrid LSTM-GRU approach has better predictive power than other approaches. Three different approaches are considered for comparison. These approaches are as follows:

1) CNN APPROACH

This approach, proposed in [56], is based on a convolutional network. A time convolution was used to apply a convolutional network to data. This network has a structure according to Table 6.

2) CNN-BILSTM-AM APPROACH

This approach was used for stock price prediction in [57]. It is a hybrid approach that uses CNN for spatial feature extraction. These spatial features are then passed to the BiLSTM layer for time series feature extraction. BiLSTM accepts the input sequence and inverse as input compared to its normal state. Finally, an attention mechanism focuses on the most related information. In this approach, the CNN

network consists of 3 consecutive layers, which are applied one-dimensionally on the stock data; in this case, it is also called temporal. This approach provides LSTM with CNN outputs that include spatial features. The LSTM used in this work is a stack that learns the time series features from the CNN output. This approach is considered a comparative approach.

3) SIMPLE LSTM APPROACH

This approach was used for stock price prediction in [58]. The model is similar to the proposed LSTM approach, with the only difference being that there is just one LSTM layer in this approach. The results obtained by different approaches are shown. Comparative approaches are trained and tested on proposed approaches in similar conditions. The amount of training data, test data, and model input parameters are the same in all these models. The CNN approach on datasets had the worst results in [56]. This indicates that this approach cannot provide essential insights into the direction of stocks. On the other hand, a CNN-BiLSTM-AM method [57] behaved similarly to the LSTM. This approach has a wide range of time-space series features. Also, the attenuation mechanism helps the model to learn better.

All experiments are run on Kaggle notebook cloud space with the following hardware:

- **CPU:** 4-core Intel Xeon
- **GPU:** NVidia K80
- **RAM:** 16GB
- **Storage:** 20GB HDD

No GPU was used in the experiments. The decision to train without a GPU was made to ensure a fair comparison between our proposed hybrid model and the baseline models evaluated by subjecting all methods to the same hardware constraints and using the same hyperparameters and technical indicators. To make a fair comparison between different models, we used the same hyperparameters and the same technical indicators for all the methods. The hyperparameter used in our experiments is illustrated in Table 7.

Table 8 shows GhMinoo and Atlas results. In GhMinoo stocks, the combined hybrid LSTM-GRU approach achieved the best results, with $MSE=0.3828$ and $MAE=0.4086$. This approach also achieved good accuracy in Atlas stocks, with $MSE=0.2474$ and $MAE=0.4696$. The LSTM approach on GhMino achieved an MSE of 0.766, compared to 0.5043 for Atlas shares. Also, GRU on these two stocks reached $MSE=0.4132$ and $MSE=0.2114$, respectively. CNN-based approaches also obtained the minimum MSE of 0.2564 in both stocks, specifically for Atlas stocks. The optimal values are written in bold.

Table 9 shows that the hybrid LSTM-GRU approach also performed better than the other approaches on Khodro data, achieving $MSE=0.2668$ and $MAE=0.2995$. Examining the MSE of the LSTM and GRU approaches shows that the LSTM approach on the Kh Motor stock has achieved

TABLE 7. Hyper parameters used in experiments.

Parameter	Value
Optimizer	Adam
Learning rate	1×10^{-3}
Number of epochs	100
Batch size	30
Number of units per module (LSTM and GRU)	50
Number of GRU modules	2
Number of LSTM modules	1

better results with an MSE of 1.2786. On the other hand, GRU achieved a better result on Qazvin with an MSE of 1.7032. The optimal values are written in bold. Table 10 shows that for D Tolid stocks, the proposed combined approach obtained worse results than the GRU and CNN-BiLSTM-AM [63] approaches. In the investigations carried out on different approaches on these two stocks, according to Table 10, it can be concluded that the combined approaches have obtained better accuracies than other approaches. This principle applies to hybrid LSTM-GRU and CNN-BiLSTM-AM models, which have obtained relatively lower MSE. The optimal values are written in bold.

Table 11 shows V BMLT and D Zahravi stocks. The proposed hybrid LSTM-GRU approach obtained worse results than the GRU and LSTM approaches. According to Table 11, it can be concluded that the behavior of these two stocks is more different from the other investigated stocks, and these two are more predictable by most approaches. The optimal values are written in bold.

Tables 12 and 13 present a comparative analysis of different models applied to various stock datasets, evaluating their performance using MSE, MAE, and MAPE. The models include LSTM, GRU, a hybrid LSTM-GRU, CNN, CNN-BiLSTM-AM, and a simple LSTM. Table 12 examines the AMZN and MSFT results, revealing that while the hybrid LSTM-GRU model does not always achieve the lowest error rates, it maintains a competitive performance. Similarly, 13 showcases the performance of the models on AAPL and IBM stocks, where the hybrid LSTM-GRU model demonstrates a balanced performance but is outperformed by other models like GRU in terms of MSE for IBM. These tables highlight that the effectiveness of each model varies depending on the specific stock. Generally, our hybrid model performs almost better than the other methods. The choice of model should be tailored to the unique characteristics of the stock being analyzed.

The proposed approaches have obtained acceptable results in each stock. The existence of low data is an essential challenge in the proposed approaches, affecting their learning. The MSE and MAE charts of different stocks for different approaches are shown in Fig. 5 and Fig. 6.

Fig. 5 compares the accuracy of different methods for all shares using MSE variables. This figure compares the accuracy of the LSTM, GRU, and hybrid LSTM-GRU methods with other methods like CNN.

TABLE 8. GhMinoo and Atlas results.

Model	Gh-Minoo			Atlas		
	MSE	MAE	MAPE	MSE	MAE	MAPE
LSTM	0.766	0.5160	294.32	0.5043	0.6626	42.393
GRU	0.4132	0.4610	256.33	0.2114	0.4272	2712
Hybrid LSTM GRU	0.3827	0.4086	166.20	0.2474	0.4695	2660
CNN[62]	0.4293	0.5323	276.20	0.3121	0.5789	22321
CNN-BiLSTM-AM [63]	0.4122	0.4678	176.44	0.2564	0.5783	2878
Simple LSTM [64]	0.4778	0.5301	295.82	0.5546	0.6890	44596

TABLE 9. Kh Motor and Qazvin results.

Model	Kh-Motor			Qazvin		
	MSE	MAE	MAPE	MSE	MAE	MAPE
LSTM	1.2786	1.0660	81.37	3.0543	1.4740	103.57
GRU	1.54	0.8821	61.189	1.7032	1.094	80.127
Hybrid LSTM GRU	2.15	0.9960	71.362	1.532	1.035	75.05
CNN[62]	1.5636	1.1234	93.12	4.043	1.5740	143.57
CNN-BiLSTM-AM [63]	2.35	1.0960	73.300	1.832	1.112	82.14
Simple LSTM [64]	1.74	0.8921	64.345	1.8931	1.934	100.329

TABLE 10. D Tolid and Khodro results.

Model	DT-Tolid			Khodro		
	MSE	MAE	MAPE	MSE	MAE	MAPE
LSTM	1.1639	0.9053	225.688	0.3697	0.4049	593.416
GRU	0.6661	0.5658	161.366	0.2861	0.3258	594.983
hybrid LSTM GRU	0.9345	0.7213	258.730	0.2668	0.2995	453.792
CNN[62]	1.631	1.0058	281.336	0.3863	0.4232	602.533
[63]CNN-BiLSTM-AM	0.7861	0.5659	163.386	0.2939	0.3434	618.123
Simple LSTM [64]	1.1712	0.9213	262.611	0.3711	0.4413	607.352

TABLE 11. V BMLT and D Zahravi results.

Model	V BMLT			D Zahravi		
	MSE	MAE	MAPE	MSE	MAE	MAPE
LSTM	0.5791	0.3113	161.772	0.2819	0.3217	250.612
GRU	0.5440	0.3212	280.761	0.1787	0.2733	197.449
Hybrid LSTM GRU	0.5456	0.3560	297.220	0.2034	0.3047	286.432
CNN[62]	0.5887	0.3203	173.702	0.2999	0.3470	253.601
CNN-BiLSTM-AM[63]	0.5801	0.3233	164.700	0.2909	0.3417	261.609
Simple LSTM [64]	0.5456	0.3590	526.298	0.2140	0.3127	298.402

TABLE 12. AMZN and MSFT results.

Model	AMZN			MSFT		
	MSE	MAE	MAPE	MSE	MAE	MAPE
LSTM	0.8891	0.6543	192.207	0.3345	0.1990	236.532
GRU	0.8152	0.5510	172.451	0.3328	0.3457	248.501
Hybrid LSTM GRU	0.8310	0.4217	131.042	0.3140	0.1990	236.532
CNN[62]	0.8987	0.7203	206.424	0.3909	0.3565	290.259
CNN-BiLSTM-AM[63]	0.9102	0.8010	219.440	0.4055	0.3722	290.752
Simple LSTM [64]	0.8327	0.4280	326.451	0.4083	0.4056	298.202

TABLE 13. AAPL and IBM results.

Model	AAPL			IBM		
	MSE	MAE	MAPE	MSE	MAE	MAPE
LSTM	0.6123	0.3357	185.215	0.3158	0.3529	275.987
GRU	0.5789	0.3471	305.872	0.2012	0.2985	210.345
Hybrid LSTM GRU	0.5892	0.3784	320.158	0.2245	0.3312	301.769
CNN[62]	0.6215	0.3468	192.541	0.3321	0.3781	280.523
CNN-BiLSTM-AM[63]	0.6154	0.3496	188.763	0.3207	0.3714	288.192
Simple LSTM [64]	0.5927	0.3298	550.921	0.2359	0.3278	312.654

Fig. 6 compares the accuracy of different methods for all shares using MAE variables. In this figure, the accuracy of the LSTM, GRU, and hybrid LSTM-GRU methods is compared with other baselines.

In the Kh Motor stock, the results obtained by LSTM are better than the proposed approach. This is due to the high volatility of this share and the lack of sufficient data

for the hybrid network. The hybrid network has about 1.5 times the learnable parameters of the LSTM and GRU networks. Moreover, in the absence of sufficient data, these learning parameters cause the model to depend too much on the data and show sensitivity; this usually causes overfitting. The same thing can be seen in the shares of D Tolid, V Melat, and D Zahravi. The GRU approach has

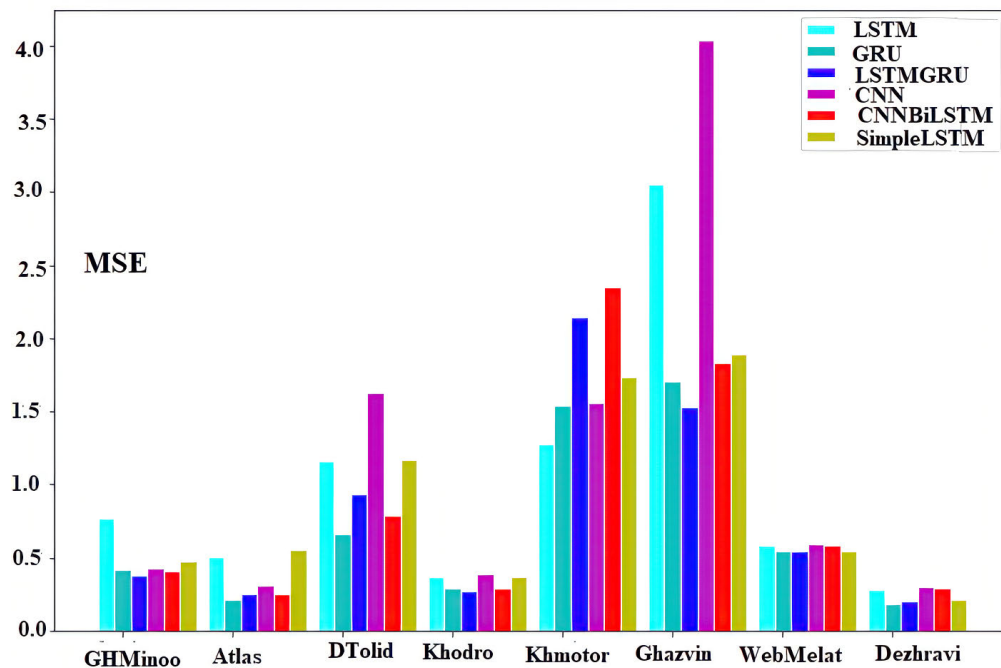


FIGURE 5. MSE chart of different stocks for different approaches.

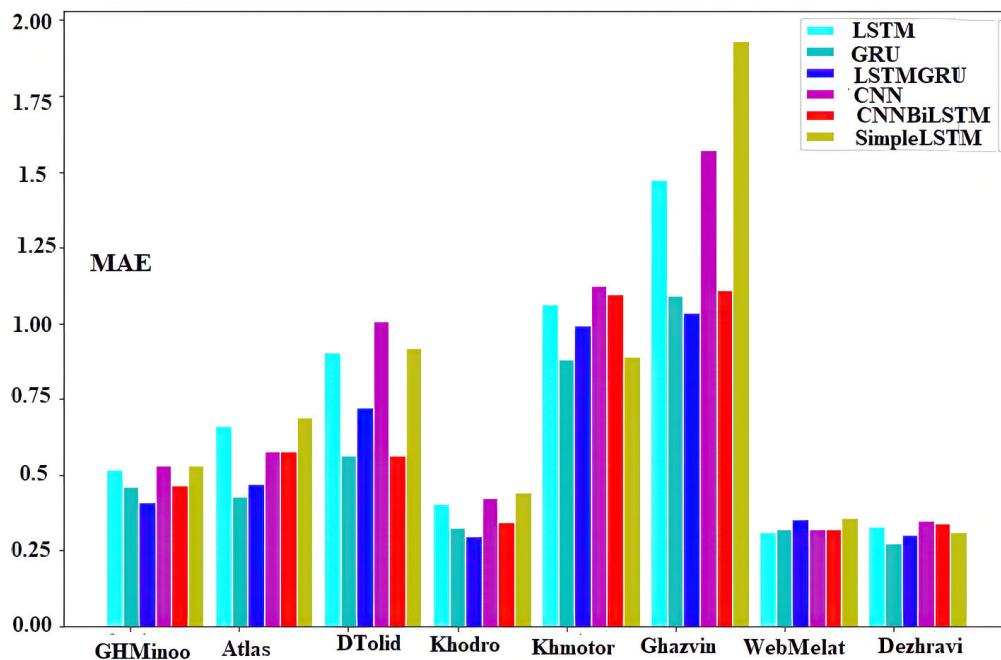


FIGURE 6. MAE chart of different stocks for different approaches.

performed better in these stocks than the proposed hybrid approach.

In Table 14, the impact of various features in different models on Atlas shares is discussed. A set of different features was considered for this case, and the results for each feature are reported. In examining the combination of different

approaches and different features, the best combination was obtained using all features. The GRU approach performed better in volatility indicators and obtained a lower MSE. This approach also obtained $MSE=0.2134$ in momentum indicators, the lowest value obtained. However, in indicators where 10 features were used, the hybrid LSTM-GRU

approach performed better than other approaches and achieved $MSE=0.2545$.

Because we have not used any particular implementation or assumption based on the stocks we have used for evaluation, we can extrapolate that the model will generalize for any other stock dataset with the input features the same as our evaluation stocks. However, for other markets like cryptocurrency that may have different characteristics or features, data-driven tests should be used to verify the generalization of the proposed method.

To ensure the reliability and robustness of our performance evaluation, all models were trained and tested using a rigorous methodology. For each stock dataset and model configuration, we conducted 10 independent experiments and averaged the results to obtain stable performance metrics. We also employed k-fold cross-validation with $k=5$ on the dataset splits to assess the models' generalization capability and minimize the risk of overfitting. The average performance metrics (MSE, MAE, and RMSE) and their standard deviations across these folds were computed and are presented in the results tables (Tables 8-13). However, standard deviations are not explicitly listed for brevity; they were calculated as part of the evaluation process. To further support the empirical findings with statistical evidence, we conducted statistical significance testing using t-tests to compare the performance of different models on the Atlas dataset, with the resulting t-statistics and p-values presented in Table 15. These tests measure confidence in the observed performance differences for this specific stock. While a comprehensive statistical analysis encompassing all pairwise model comparisons across every evaluated stock was beyond the scope of the current study, the use of multiple runs, k-fold cross-validation, and the presented statistical test results for Atlas provide empirical support for the performance observations. To further validate the robustness of our comparative evaluation and ascertain whether the observed performance differences between models are statistically significant, we conducted independent t-tests comparing the mean MSE obtained by each model based on the results from our k-fold cross-validation approach. Table 15 presents the t-statistic and p-value for pairwise comparisons of the LSTM, GRU, hybrid LSTM-GRU, CNN, CNN-BiLSTM-AM, and Simple LSTM models specifically for the Atlas stock dataset, building upon the detailed analysis shown in Table 14. A low p-value (typically below 0.05) indicates that the difference in performance between the two models is statistically significant, suggesting that the observed difference is unlikely due to random chance. The results in Table 15 allow us to draw more reliable conclusions about the relative effectiveness of the models, providing statistical support for the performance trends discussed. Addressing the practical implementation of the hybrid LSTM-GRU model for stock market prediction necessitates a discussion of computational considerations. As noted, our experiments were conducted in a CPU-only environment, which provides a baseline for understanding the model's demands. The

complexity of deep learning architectures like our proposed hybrid LSTM-GRU model demands substantial processing power, memory, and training time. While we performed 10 experiments for each stock and averaged the results using this setup, the training duration for the full model on large datasets is considerable, posing challenges for rapid iteration or deployment on resource-constrained hardware. The hybrid architecture, having approximately 1.5 times the learnable parameters of individual LSTM or GRU networks, contributes to this computational cost. Furthermore, scaling the model to handle ever-increasing datasets or a wider range of stocks requires architectural optimization and potentially parallel processing techniques. Maintaining prediction performance with continuous data ingestion also necessitates regular model updates and retraining, adding to the ongoing computational burden. These computational costs and the need for efficient processing of continuous data streams present significant challenges for near-real-time deployment. Future work must explore more efficient model architectures, optimization techniques, or hardware acceleration (such as GPUs) to mitigate these limitations and enhance the model's real-world applicability in time-sensitive financial scenarios. A crucial aspect of deploying sophisticated deep learning models like the hybrid LSTM-GRU for financial forecasting is the challenge of model interpretability. As noted previously, these architectures' complex, non-linear nature often results in a "black box" effect, making it difficult to trace precisely why the model arrives at a specific prediction. This lack of transparency can be a significant concern for financial stakeholders, including investors, analysts, and regulatory bodies, who often require explainable decisions rather than just accurate predictions. Addressing this interpretability gap is a key area for future work and responsible model deployment. Complementary to interpretability, understanding the influence of input features, particularly the technical indicators, on the model's output is vital. Our approach leverages a rich feature space comprising 42 technical indicators derived from historical price and volume data and the main features. These indicators are designed to capture various market trends, momentum, and volatility signals. Our experimental results, particularly those presented in Table 14, implicitly demonstrate their significant influence; using the comprehensive set of indicators alongside historical data yields better prediction performance than using only subsets of these features. This underscores the importance of selecting and utilizing a rich and relevant set of technical indicators for effective stock market prediction with deep learning models. Although empirical results show that model performance can vary depending on the specific stock being analyzed, our evaluation across eight distinct TSE stocks, using multiple experimental runs and k-fold cross-validation, demonstrates that the proposed hybrid model achieves competitive or generally improved predictive accuracy when compared to several baseline models on these datasets. Statistical significance testing for comparisons on the Atlas stock provides further evidence for the observed

TABLE 14. The results obtained for Atlas by different feature sets.

Features	Dimension	Model	MSE	MAE	MAPE
Indicators	R^{10}	LSTM	0.5141	0.6823	45393
		GRU	0.2633	0.4972	2913
		Hybrid LSTM-GRU	0.2545	0.4891	2861
Momentum Indicators	R^{25}	LSTM	0.5242	0.6924	45892
		GRU	0.2134	0.4775	2853
		Hybrid LSTM-GRU	0.2442	0.4796	2761
Volatility Indicators	R^{30}	LSTM	0.5342	0.6898	45878
		GRU	0.2234	0.4634	2834
		Hybrid LSTM-GRU	0.2542	0.4709	2883
All features (Indicators, Momentum, Volatility, Historical Data)	R^{42}	LSTM	0.5043	0.6626	42393
		GRU	0.2114	0.4272	2712
		Hybrid LSTM-GRU	0.2474	0.4695	2660

TABLE 15. T-stat and P-value for different models on Atlas.

	LSTM	GRU	hybrid LSTM-GRU	CNN [62]	CNN [63]	BiLSTM-AM	Simple LSTM [64]
LSTM	tstat= 0.0000 pvalue= 1.0000	tstat= 0.9341 pvalue= 0.403113	tstat= 0.9354 pvalue= 0.402521	tstat= 0.4190 pvalue= 0.696763	tstat= 0.9300 pvalue= 0.405018		tstat= -0.0358 pvalue= 0.973153
GRU	tstat= -0.9341 pvalue= 0.403113	tstat= 0.0000 pvalue= 1.0000	tstat= 0.0137 pvalue= 0.989748	tstat= -0.8721 pvalue= 0.432366	tstat= -0.0420 pvalue= 0.968487		tstat= -0.9375 pvalue= 0.401587
Hybrid LSTM-GRU	tstat= -0.9354 pvalue= 0.402521	tstat= -0.0137 pvalue= 0.989748	tstat= 0.0000 pvalue= 1.0000	tstat= -0.8747 pvalue= 0.431130	tstat= -0.0557 pvalue= 0.958279		tstat= -0.9387 pvalue= 0.401027
CNN [62]	tstat= -0.4190 pvalue= 0.696763	tstat= 0.8721 pvalue= 0.432366	tstat= 0.8747 pvalue= 0.431130	tstat= 0.0000 pvalue= 1.0000	tstat= 0.836 pvalue= 0.436353		tstat= -0.4467 pvalue= 0.678227
CNN BiLSTM-AM [63]	tstat= -0.9330 pvalue= 0.405018	tstat= 0.0420 pvalue= 0.968487	tstat= 0.0557 pvalue= 0.958279	tstat= -0.8639 pvalue= 0.436353	tstat=0.0000 pvalue= 1.0000		tstat= -0.9335 pvalue= 0.403387
Simple LSTM [64]	tstat= -0.0358 pvalue= 0.973153	tstat= 0.937 pvalue= 0.401587	tstat= 0.9387 pvalue= 0.401027	tstat= 0.4467 pvalue= 0.678227	tstat= 0.9335 pvalue= 0.403387		tstat= 0.0000 pvalue= 1.0000

performance differences. Thus, this paper's contributions include offering a rigorous analysis of a specific hybrid deep learning approach tailored with extensive technical indicators for stock prediction in the context of a less-studied market.

VI. CONCLUSION

In our work on stock price prediction, we have developed an approach using hybrid LSTM-GRU models. These models are popular for their ability to learn from sequential data, and our architecture is uniquely structured with GRU layers processing initial features before LSTM layers analyze more complex patterns. This design aims to efficiently capture both short-term and long-term dependencies in stock data while simplifying the often complex hyperparameter tuning process. GRU layers are known for their computational efficiency and effectiveness in capturing short- and medium-term temporal dependencies. By positioning GRU layers at the initial stage, the model can efficiently extract relevant patterns from the input data while also contributing to mitigating the vanishing gradient problem, facilitating better gradient flow for initial feature learning. The output from the GRU layers then serves as input to the LSTM layers. LSTM

networks, equipped with more intricate gating mechanisms, are particularly adept at capturing long-term dependencies. This sequential structure allows the LSTM layers to refine the features learned by the GRUs and specifically focus on the long-term trends and dependencies crucial for accurate stock price forecasting, which is inherently influenced by short-term fluctuations and prevailing long-term market trends. This arrangement enables the proposed hybrid model to capture this dual temporal nature of stock market data effectively. While alternative architectures, such as bidirectional or parallel arrangements of GRU and LSTM layers or models incorporating attention mechanisms, present interesting possibilities and have shown benefits in related work, these were not explored within the scope of the current hybrid model's design presented in this study. Nevertheless, these alternative architectures and the integration of attention mechanisms are important and promising avenues for future research to enhance prediction performance further. Beyond the model, we have significantly enhanced the input data by carefully engineering a comprehensive set of features. We have gone beyond basic price information to include 42 technical indicators, covering momentum, volatility, and

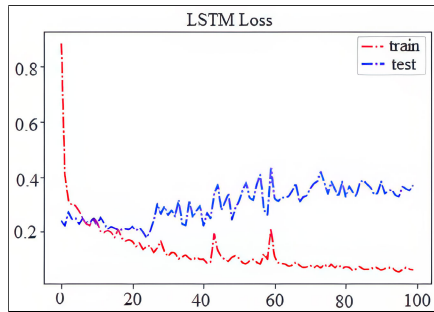


FIGURE 7. Khodro LSTM loss.

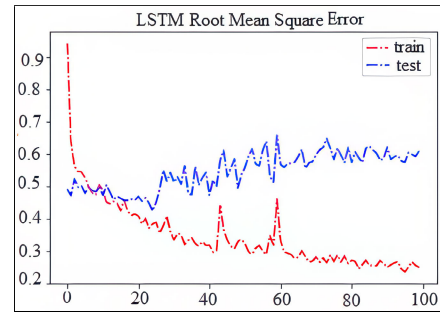


FIGURE 10. Khodro LSTM RMSE.

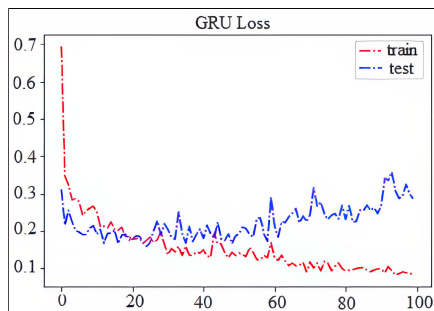


FIGURE 8. Khodro GRU loss.

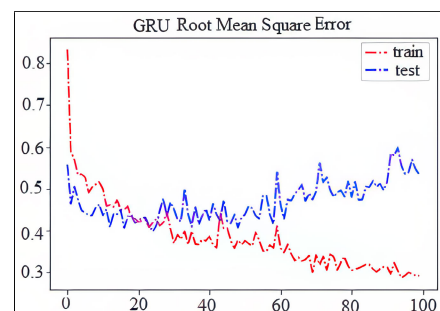


FIGURE 11. Khodro GRU RMSE.

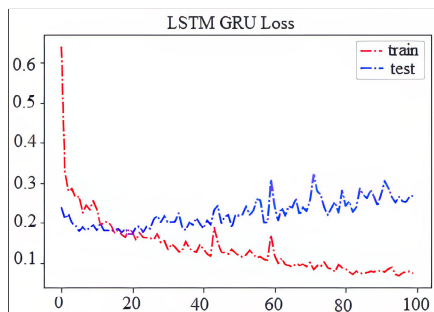


FIGURE 9. Khodr hybrid loss.

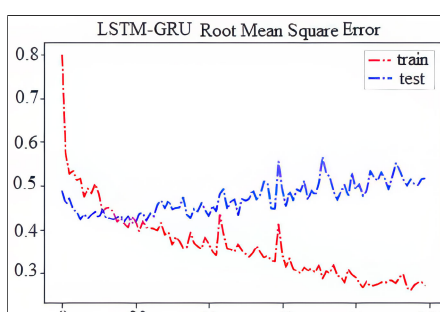


FIGURE 12. Khodro hybrid RMSE.

trend analysis derived from the Stockstats library. This extensive feature set is a key contribution, as prior research has not established a definitive set of technical indicators for stock prediction. Finally, our study offers specific, market-focused insights by concentrating on stock price prediction within the Tehran Stock Exchange (TSE). We have analyzed a decade of data for eight different TSE stocks, and our results provide valuable observations about how our hybrid model and other methods perform in this specific market, highlighting variations in performance and potential influencing factors like stock volatility and data availability.

This research does not include essential information such as unemployment rate, sales, GDP, etc. It is solely a technical analysis based on the actual price and aimed to predict the Tehran Stock Exchange stock market based on deep learning methods. Suggested strategies for investment are designed for short-term investments. According to the diversity, the suggested strategies for deep learning algorithms stock can

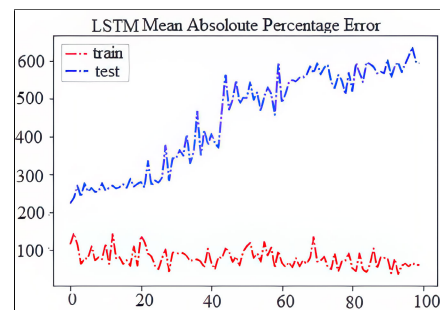


FIGURE 13. Khodro LSTM MAPE.

work well in these predictions. Three different approaches were proposed: LSTM, GRU, and hybrid LSTM-GRU. The proposed hybrid LSTM-GRU approach had a lower error rate than the other approaches, according to the results obtained from the tables. Deep reinforcement learning approaches

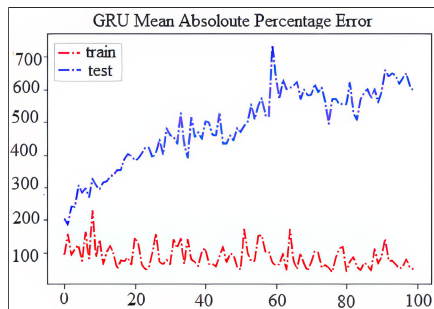


FIGURE 14. Khodro GRU MAPE.

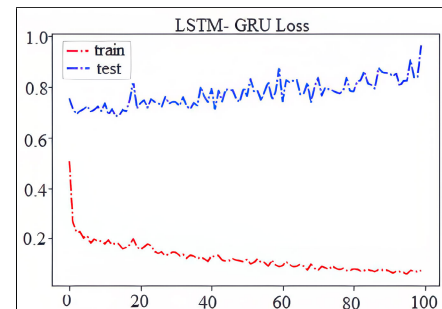


FIGURE 18. GH-Minoo hybrid loss.

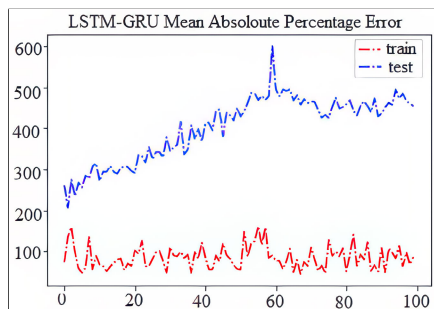


FIGURE 15. Khodro hybrid MAPE.

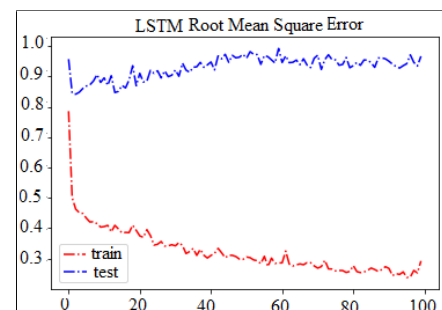


FIGURE 19. GH-Minoo LSTM RMSE.

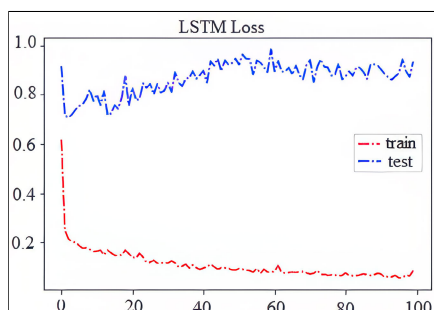


FIGURE 16. GH-Minoo LSTM loss.

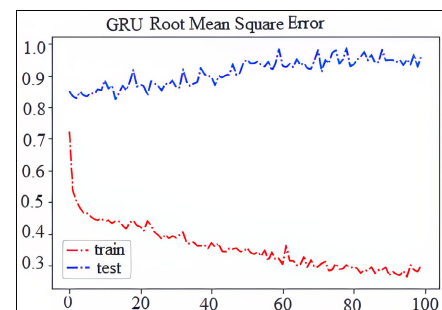


FIGURE 20. GH-Minoo GRU RMSE.

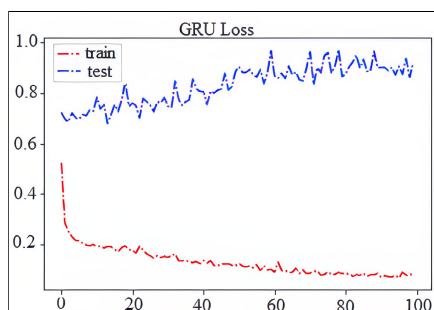


FIGURE 17. GH-Minoo GRU loss.

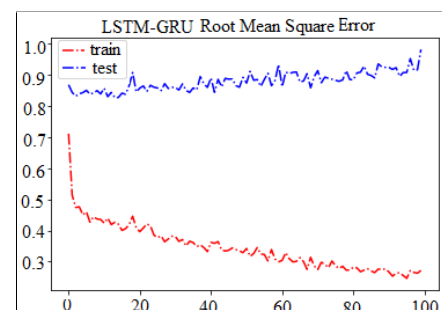


FIGURE 21. GH-Minoo Hybrid RMSE.

have also achieved acceptable results in some cases. The use of deep reinforcement learning approaches is one of the examples that can be implemented on this data in the future. RNN approaches applied to the dataset were able to achieve appropriate accuracy. Combining these approaches, it is concluded that using heterogeneous layers (networks with

different recurrent layers) can lead to better learning. In the conducted surveys, it was observed that some models are overfitting the data, which can be a weak point of these models. Furthermore, using AI-driven trading models, such as the proposed hybrid LSTM-GRU network, raises ethical issues

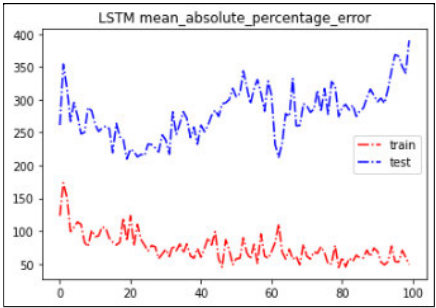


FIGURE 22. GH-Minoo LSTM MAPE.

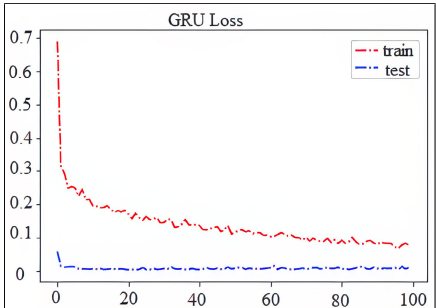


FIGURE 26. Qazvin GRU loss.

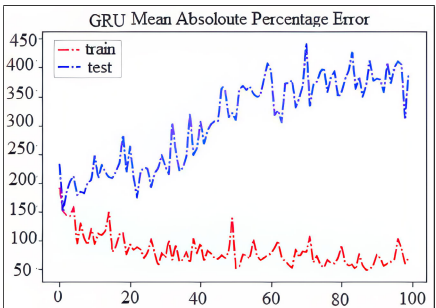


FIGURE 23. GH-Minoo GRU MAPE.

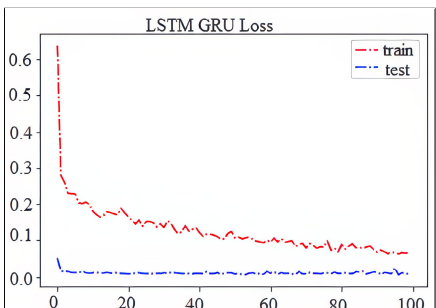


FIGURE 27. Qazvin hybrid loss.

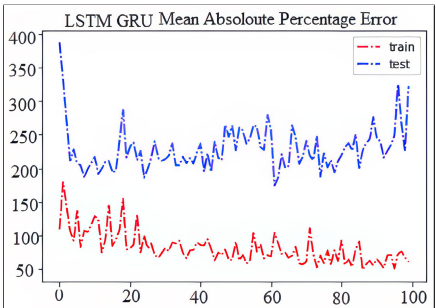


FIGURE 24. GH-Minoo hybrid MAPE.

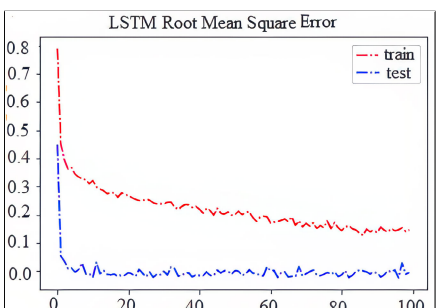


FIGURE 28. Qazvin LSTM RMSE.

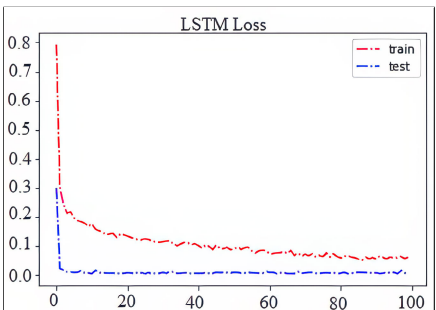


FIGURE 25. Qazvin LSTM loss.

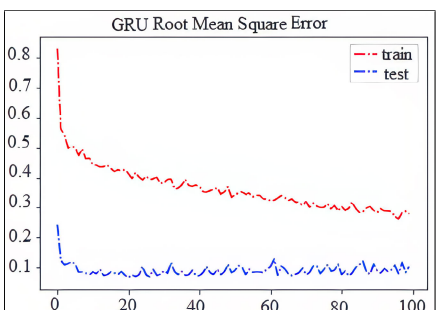


FIGURE 29. Qazvin GRU RMSE.

concerning market manipulation and stability. Automated systems operating at high speeds could potentially contribute to rapid market shifts or exacerbate volatility, demanding careful implementation and monitoring to prevent unintended consequences. Another crucial ethical consideration is the

potential for algorithmic bias. As these models learn from historical market data, biases inherent in that data (e.g., reflecting past market inefficiencies or specific participant behaviors) could be learned and perpetuated by the algorithm. This could lead to unfair or skewed trading outcomes for

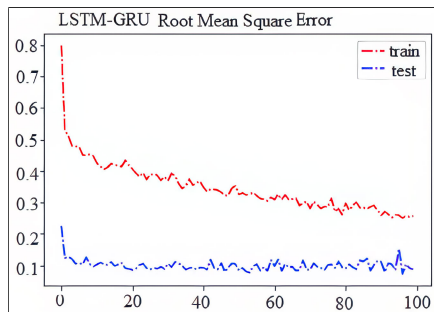


FIGURE 30. Qazvin hybrid RMSE.

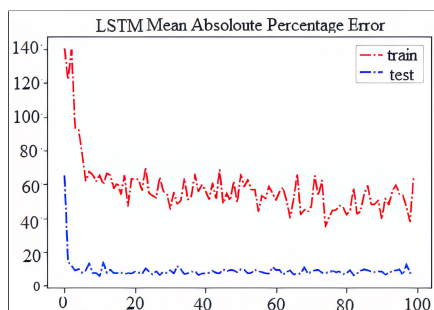


FIGURE 31. Qazvin LSTM MAPE.

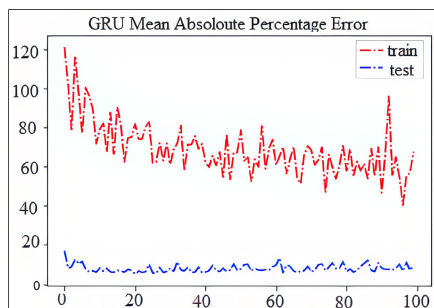


FIGURE 32. Qazvin GRU MAPE.

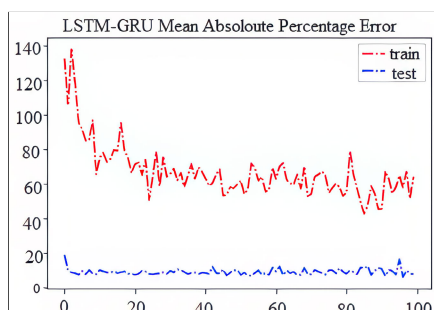


FIGURE 33. Qazvin hybrid MAPE.

different assets or market participants. Addressing these risks through responsible model development, transparency, and robust regulatory frameworks is essential for ethically deploying such advanced trading systems in real-world financial markets. Furthermore, this study does not include

practical financial evaluations using trading-relevant metrics such as profitability, Sharpe ratio, or cumulative returns, as the available dataset from the Tehran Stock Exchange lacks essential variables (like transaction costs, bid/ask spreads, or a reliable risk-free rate) required for their accurate calculation. Future research will explore several avenues to enhance stock market prediction accuracy. This includes testing other hybrid models on stock market data. Another promising direction is expanding the feature space by incorporating additional financial indicators, such as macroeconomic variables, market sentiment, and geopolitical events. These factors could provide a more holistic view of market influences. Given the success of transformer-based models like Transformer and BERT in other domains, integrating these architectures with the current hybrid LSTM-GRU approach will also be investigated, aiming to leverage their ability to capture complex dependencies. Finally, future work will address the practical implementation challenges of the hybrid model, focusing on computational cost, scalability, and interpretability to facilitate real-world applications. These efforts aim to advance the field of accurate and efficient stock market prediction. Future research will explore several avenues to enhance stock market prediction accuracy. Building upon the current work, expanding the feature space by incorporating additional data streams, such as macroeconomic variables, geopolitical events, and market sentiment derived from sources like news or social media, represents a promising direction. This approach, effectively a form of multi-modal data fusion, aims to capture a more holistic view of market dynamics beyond purely technical indicators. Furthermore, given the significant advancements of transformer-based architectures like Transformer and BERT in sequence modeling, investigating the integration of these models with or as alternatives to the current hybrid LSTM-GRU approach will be a key focus. This could potentially leverage their advanced attention mechanisms to capture complex dependencies better. Finally, addressing the identified practical implementation challenges, such as the model's computational cost and scalability, is essential for real-world applicability. Future efforts will include exploring more efficient architectures or optimization techniques, potentially leading to the development of lightweight models suitable for deployment in resource-constrained environments, such as mobile trading applications.

Although the hybrid LSTM-GRU model shows promise for stock market prediction, its limitations, and practical implications must be considered. Model performance depends heavily on the quality and availability of historical data, and market shifts, unexpected events, and changing investor behavior can negatively affect accuracy. Despite overfitting mitigation efforts, the risk persists, necessitating regular evaluation and retraining. Given the inherent volatility and unpredictability of the stock market, consistent, accurate predictions remain difficult, and financial losses are always possible. Furthermore, using AI-driven trading models, including this one, raises ethical issues concerning market

manipulation and stability. Future model enhancements could include incorporating attention mechanisms to focus on relevant input data, exploring transformer-based architectures adapted from natural language processing, and potentially combining predictions from multiple models, including the proposed one, to improve overall accuracy and robustness.

APPENDIX

1. The detailed training plots for different stocks:

See Figs. 7–33.

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